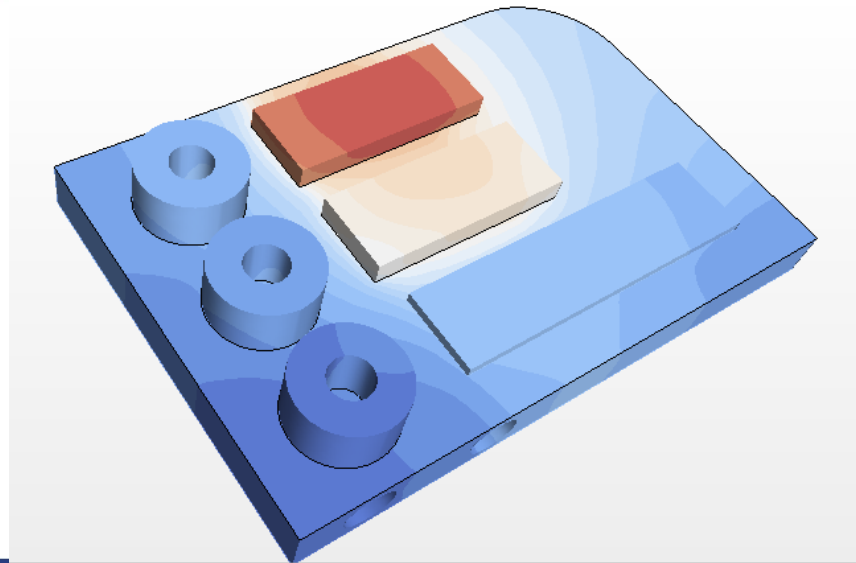




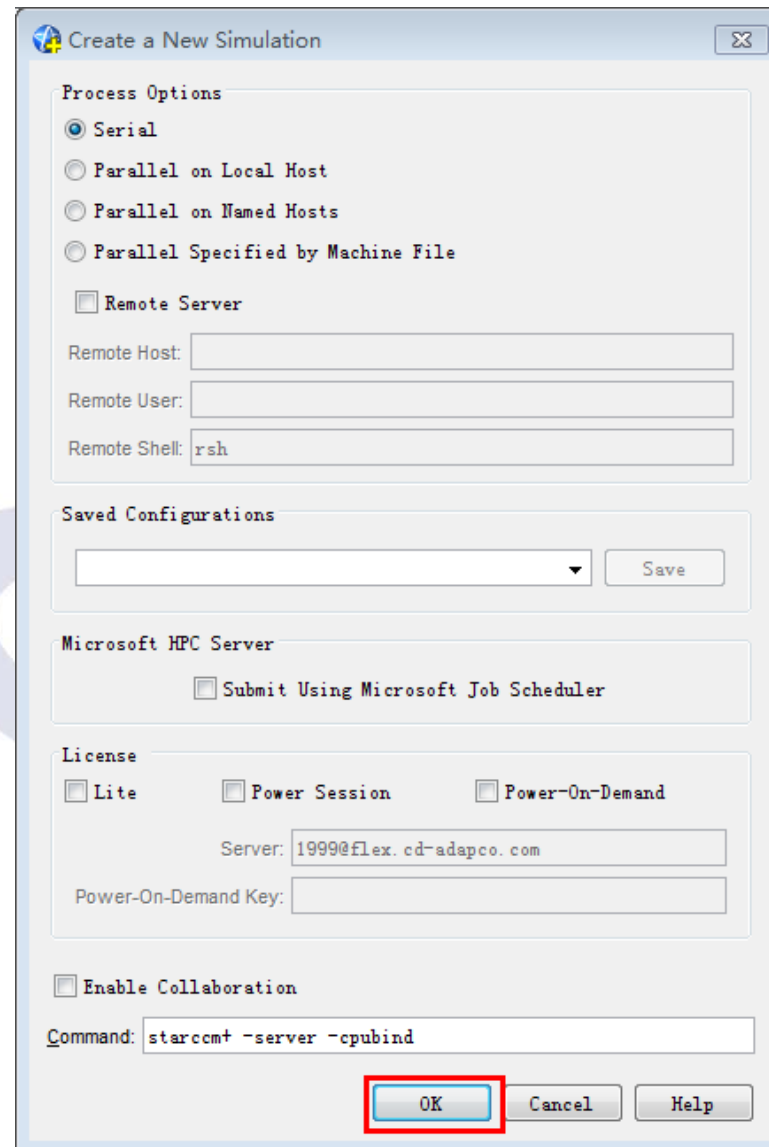
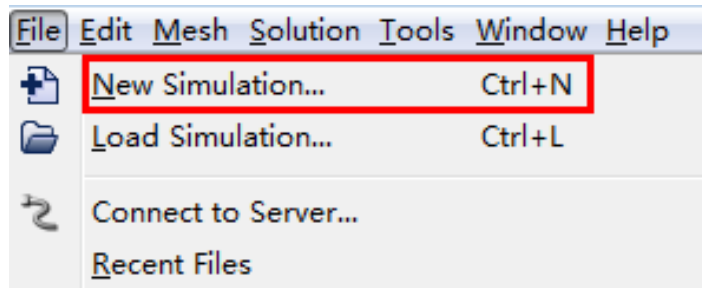
Workshop2 CHT解析概要

概要

- 介绍热传导，热对流以及热辐射的设定
 - CHT的Imprint, Meshing以及Interface设定
 - 壁面和近壁面的处理
 - 发热定义
 - 接触热阻
 - S2S热辐射
 - 热边界条件
 - 热传导系数

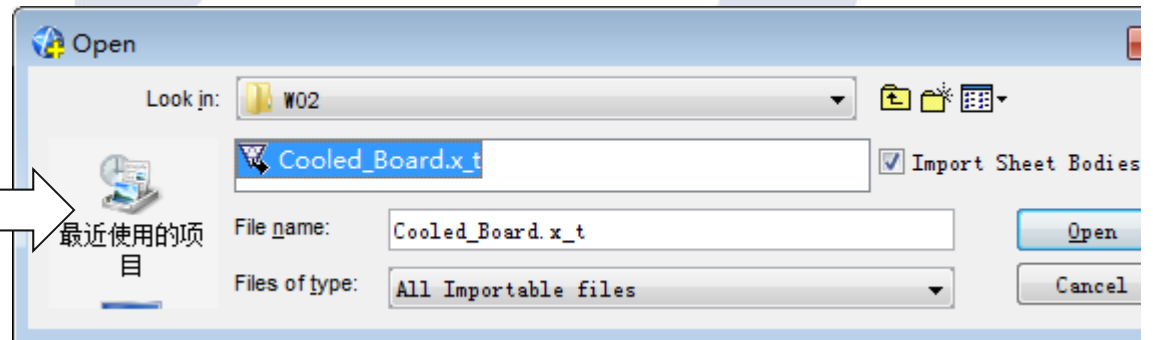
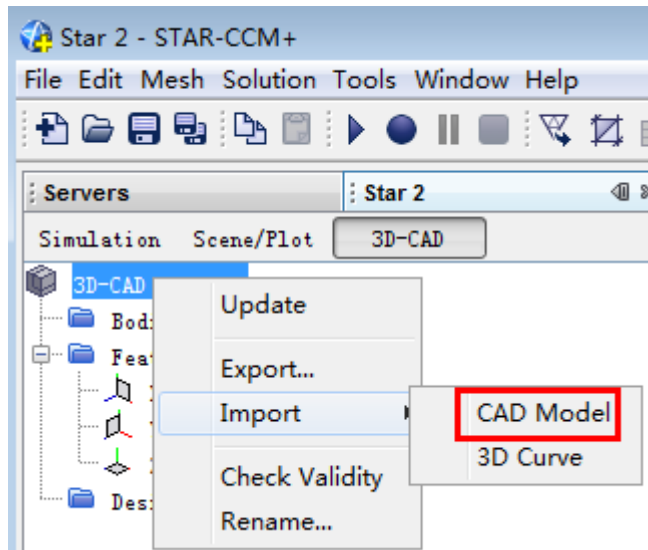
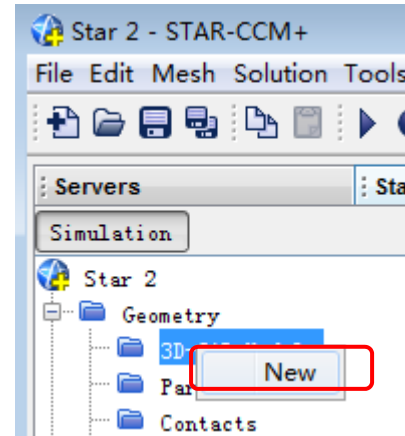
CHT Demo (新建模拟)

1. 启动STAR-CCM+。
2. [File] > [New Simulation] > [OK]



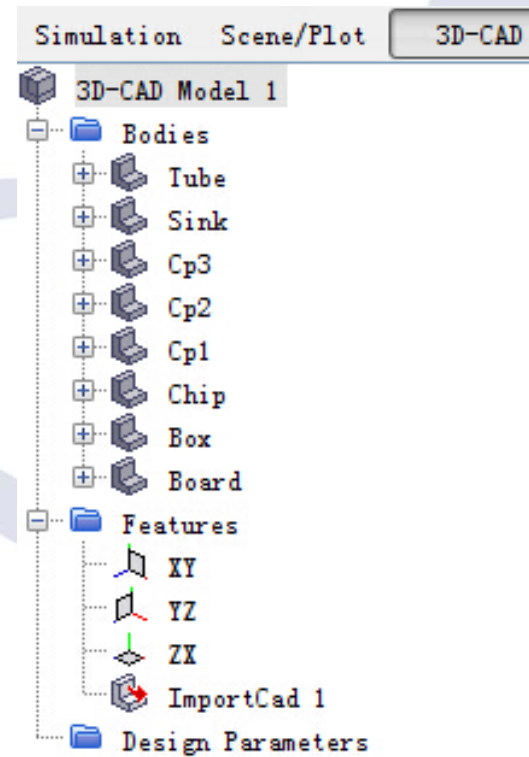
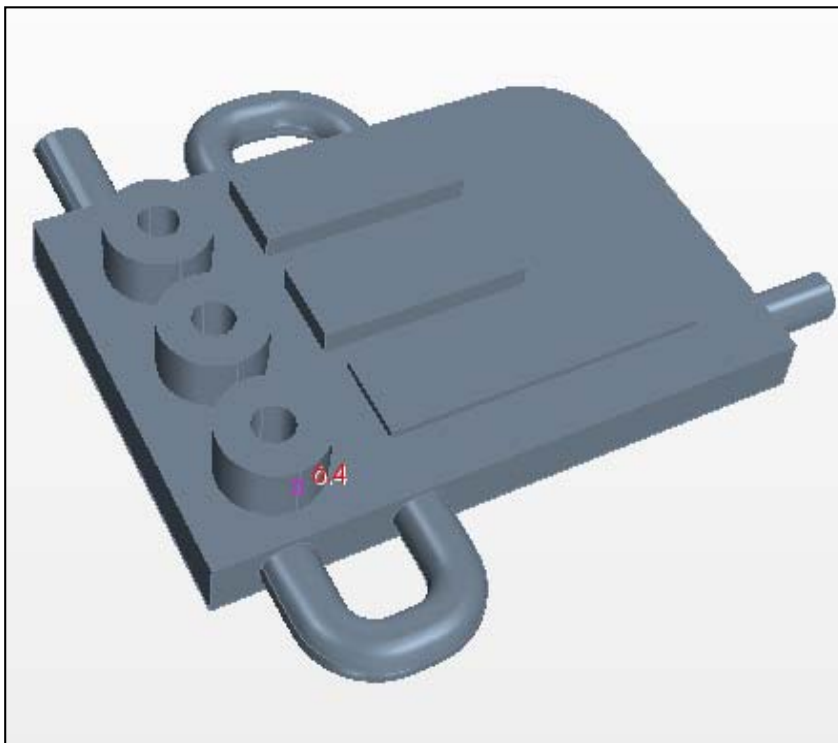
CHT Demo (导入CAD)

1. [Geometry] > [3D-CAD Models] > [New]
2. [3D-CAD Model 1] > [Import] > [CAD Model]
3. 选择Cooled_Board.x_t。



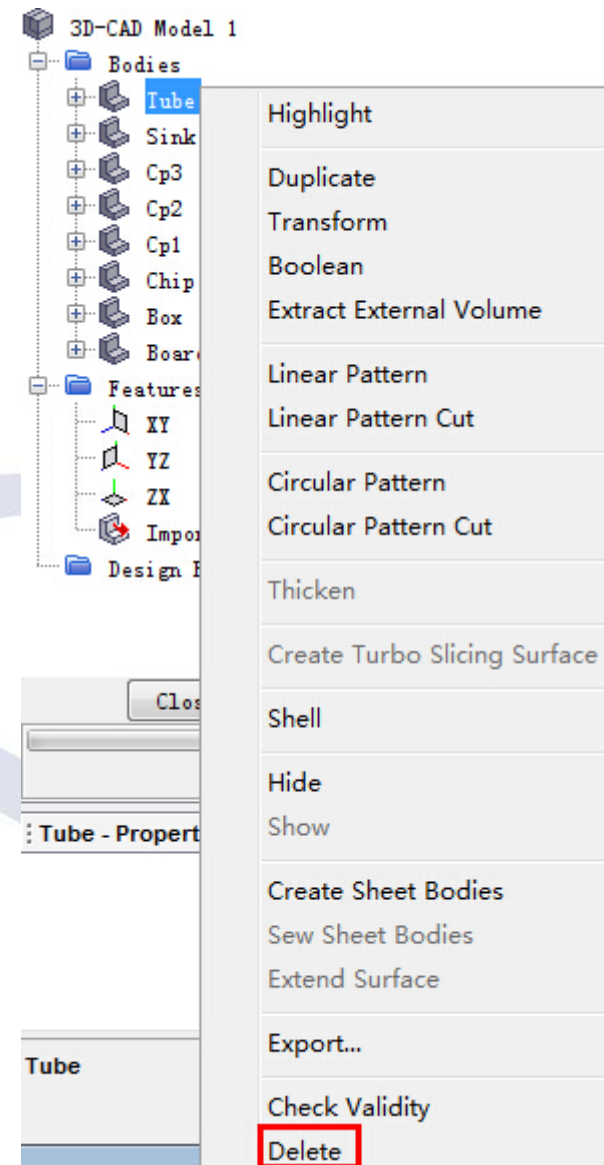
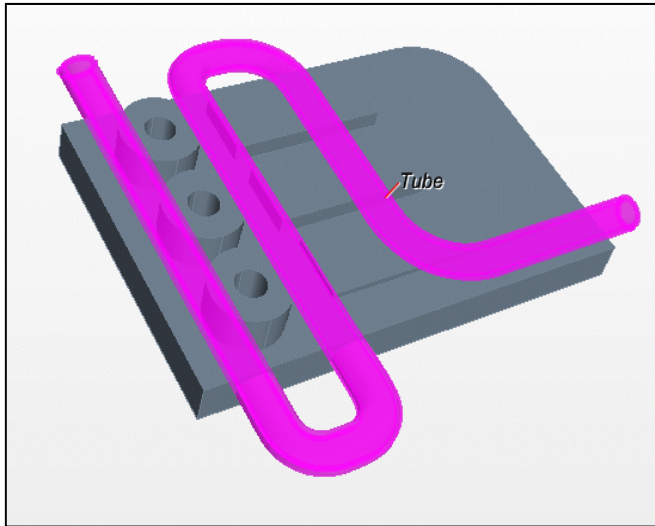
CHT Demo (检查CAD Model)

- 下图为导入的几何，在本体上有若干构件，中间管道内流动介质为冷却水
 - 需要删除冷却管路，生成周围的空间



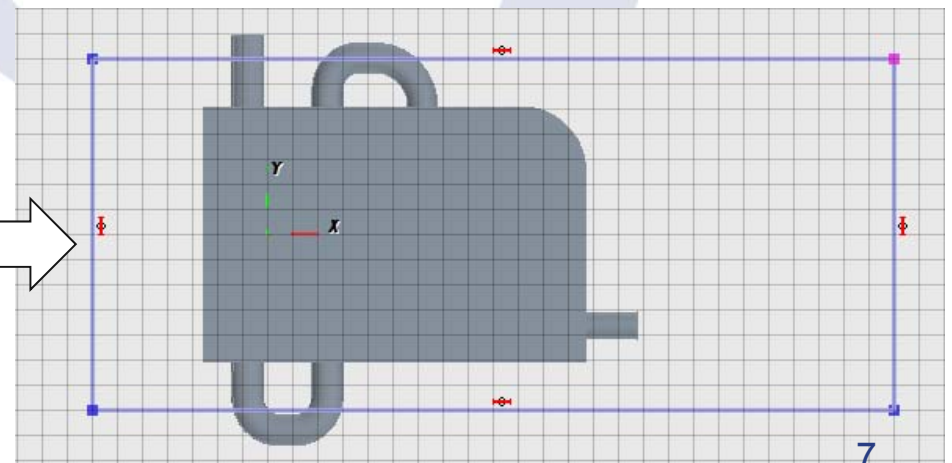
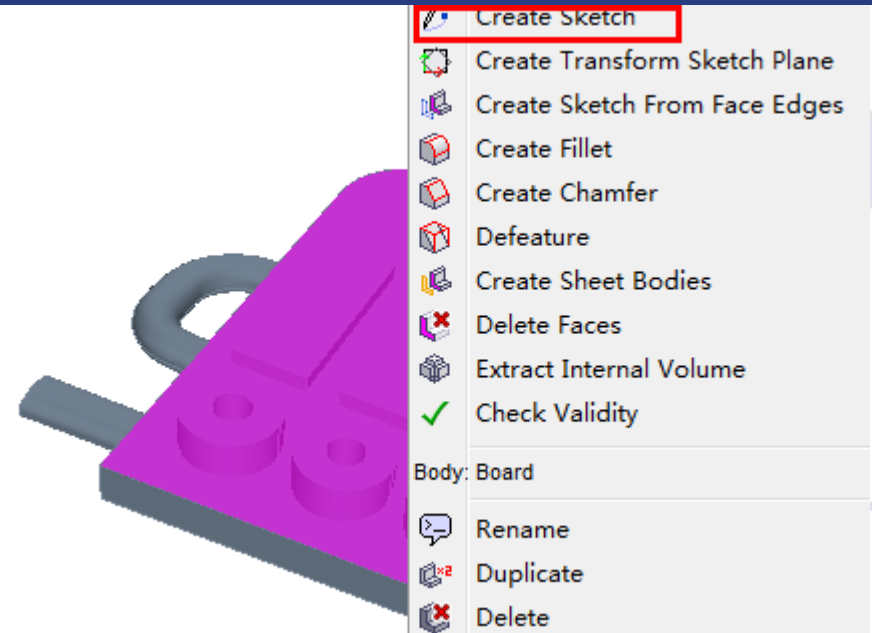
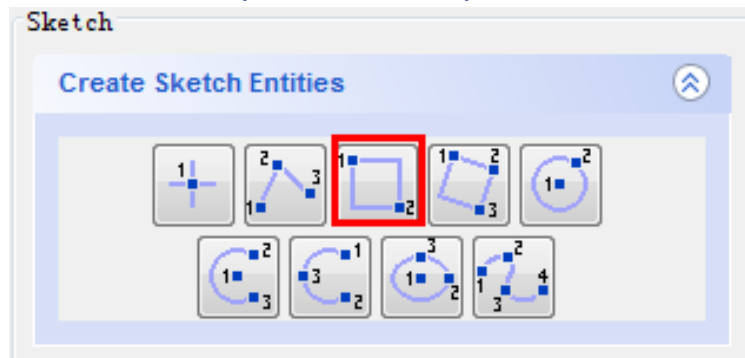
CHT Demo (删除冷却管路)

1. 右击[Bodies] > [Tube], 选择 [Delete]
 - 冷却管路被删除。



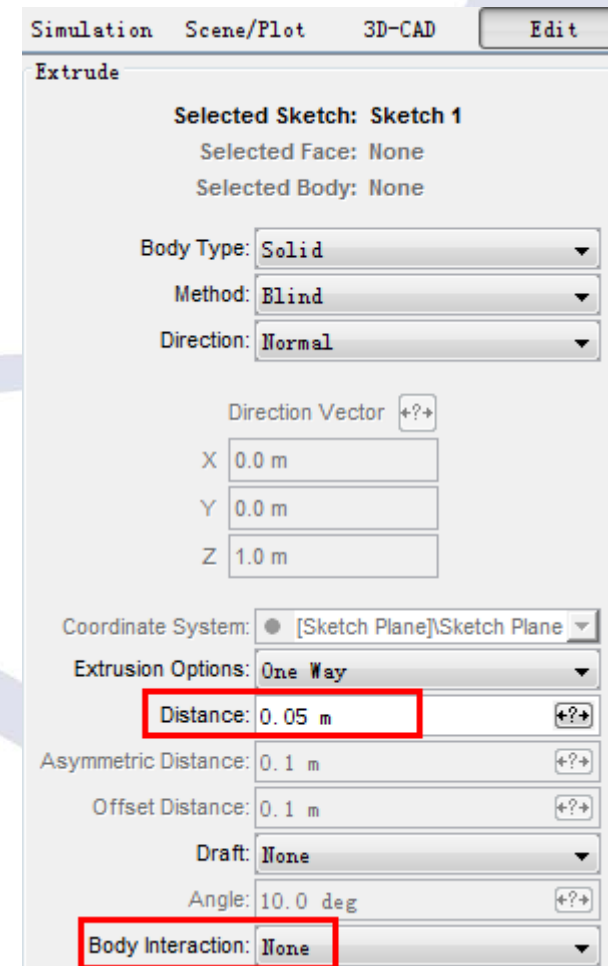
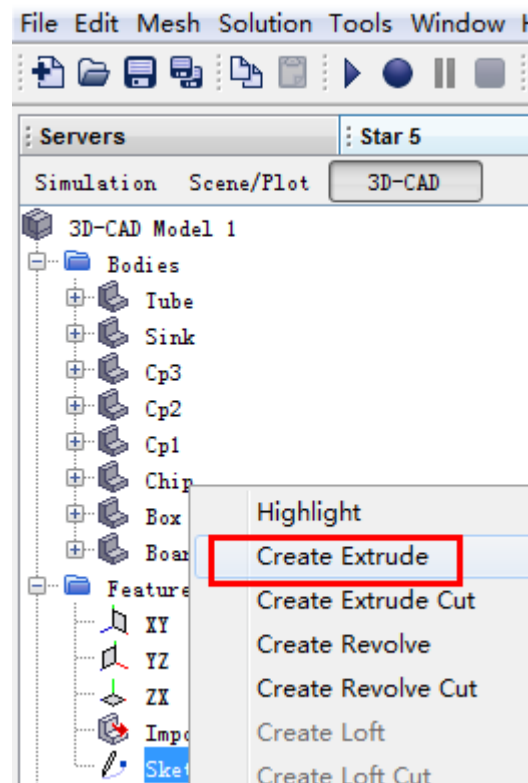
CHT Demo (绘制Sketch)

1. 在图示平面上右击，选择[Create Sketch]
2. 在Sketch窗口上，选择[Create rectangle]图标，指定以下点的坐标
 - 左下(-0.07, -0.07)
 - 右上(0.25, 0.07)



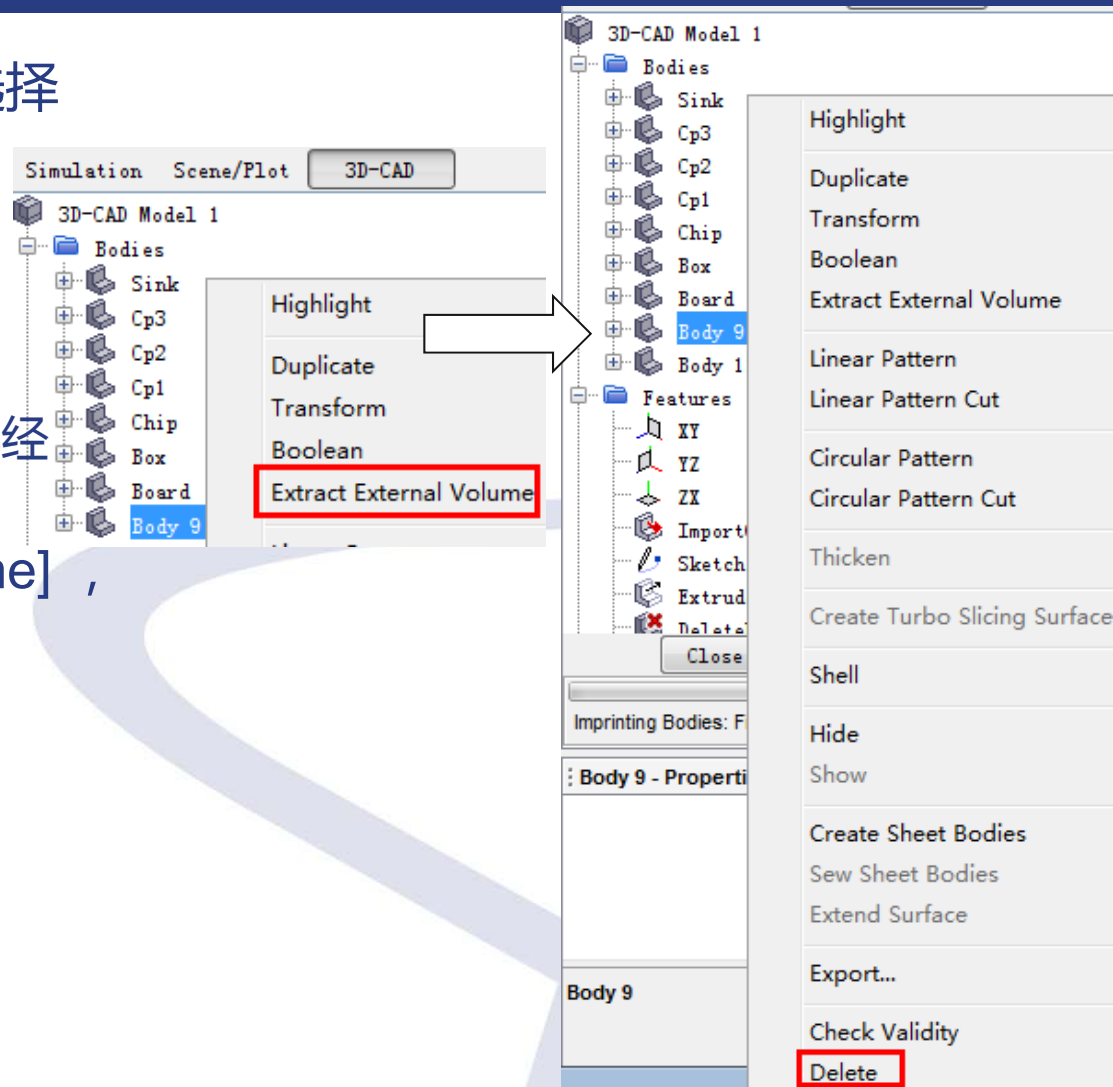
CHT Demo (生成拉伸体)

1. 右击[Features] > [Sketch 1] , 选择[Create Extrude]
2. 指定拉伸"Distance"为0.05m
3. Body Interaction选择"None"



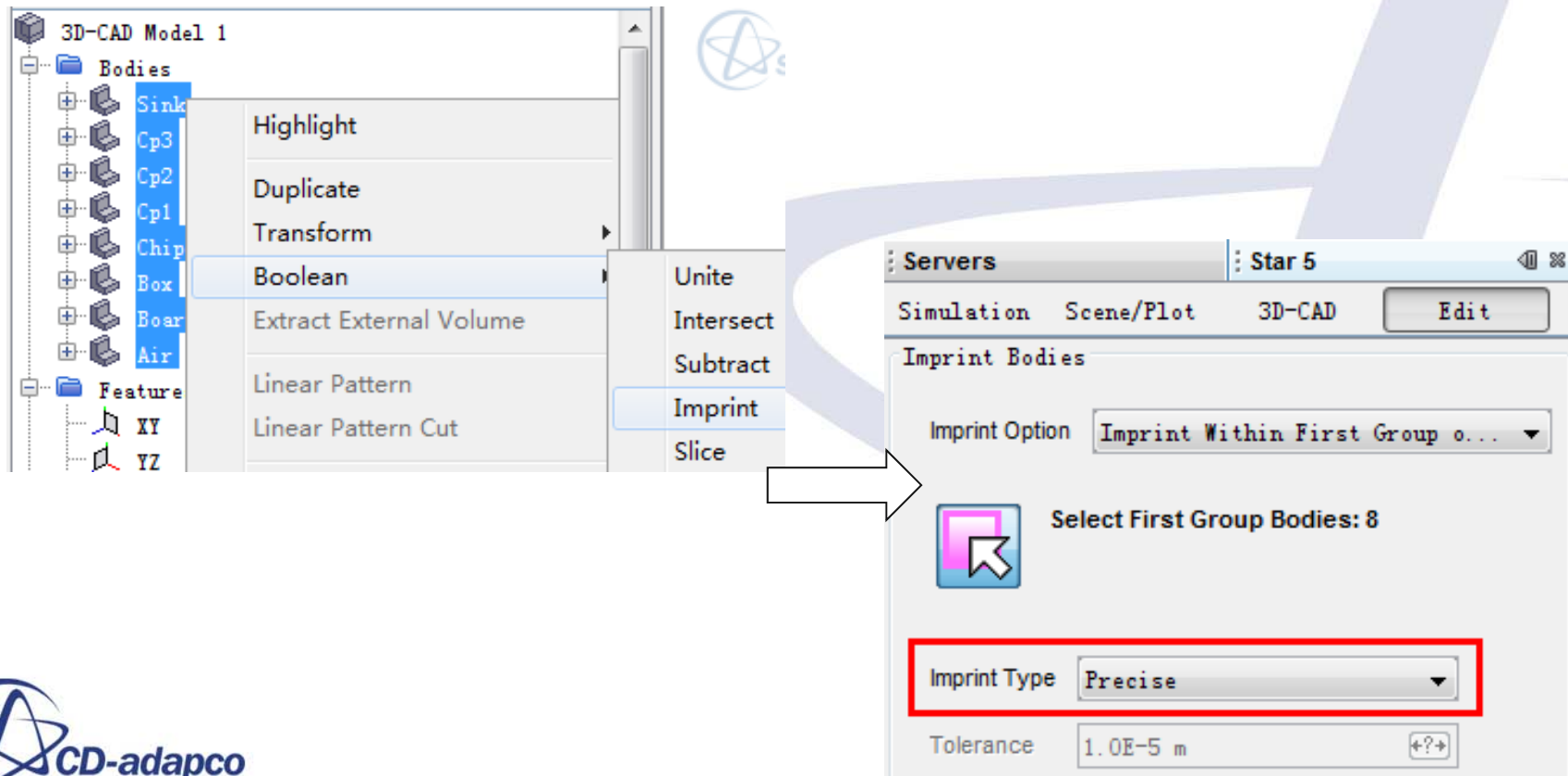
CHT Demo (抽取外部空间以及部件删除)

1. 右击[Bodies] > [Body 9] , 选择 [Extract External Volume]
 - 抽出新的流体域。
 - 新抽出的流体域为Body 10。
2. 右击[Body 9] , 选择 [Delete]
 - 因为已经抽出了外部空间, 已经不再需要Body 9了
3. 右击[Body 10] , 选择[Rename] , 将名称改为Air



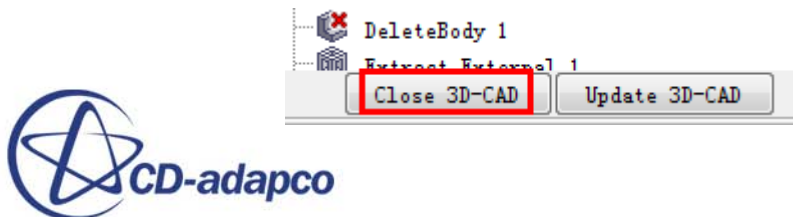
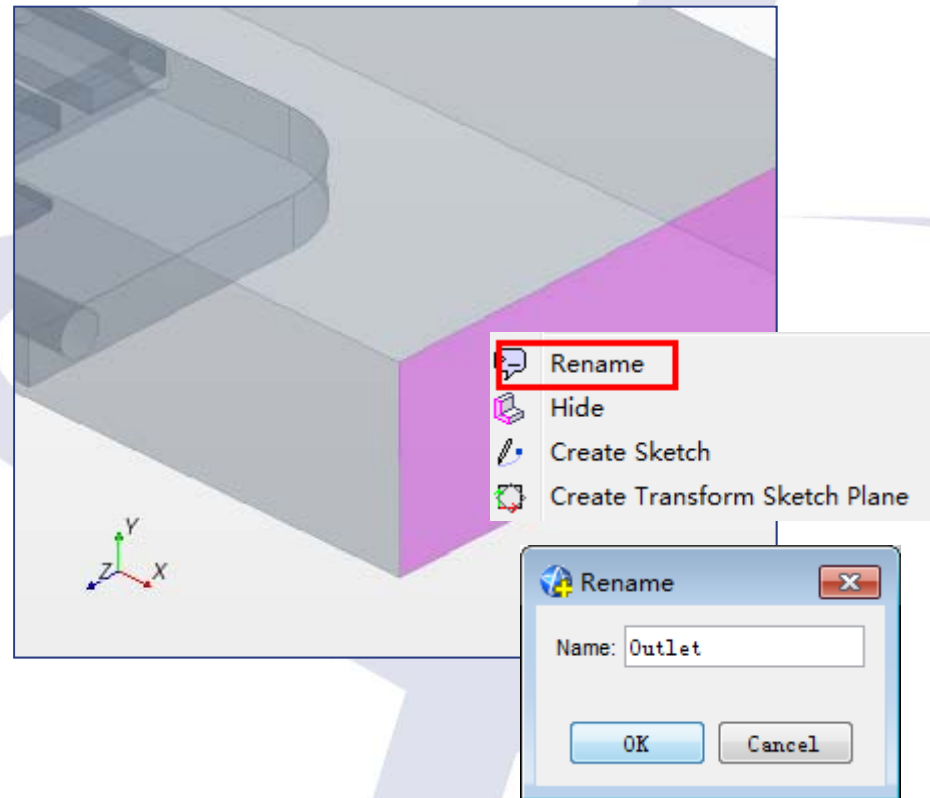
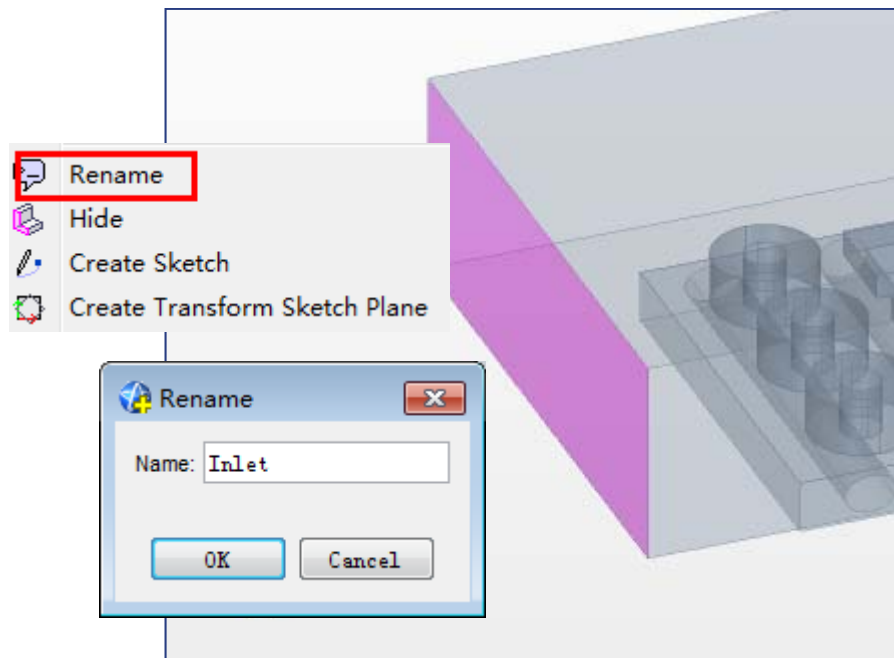
CHT Demo (部件间压印)

1. 选择所有的部件，右击，选择[Boolean] > [Imprint]
 - 为了生成共形网格，需要在部件之间实施压印操作。
2. Imprint Type选择“Precise”



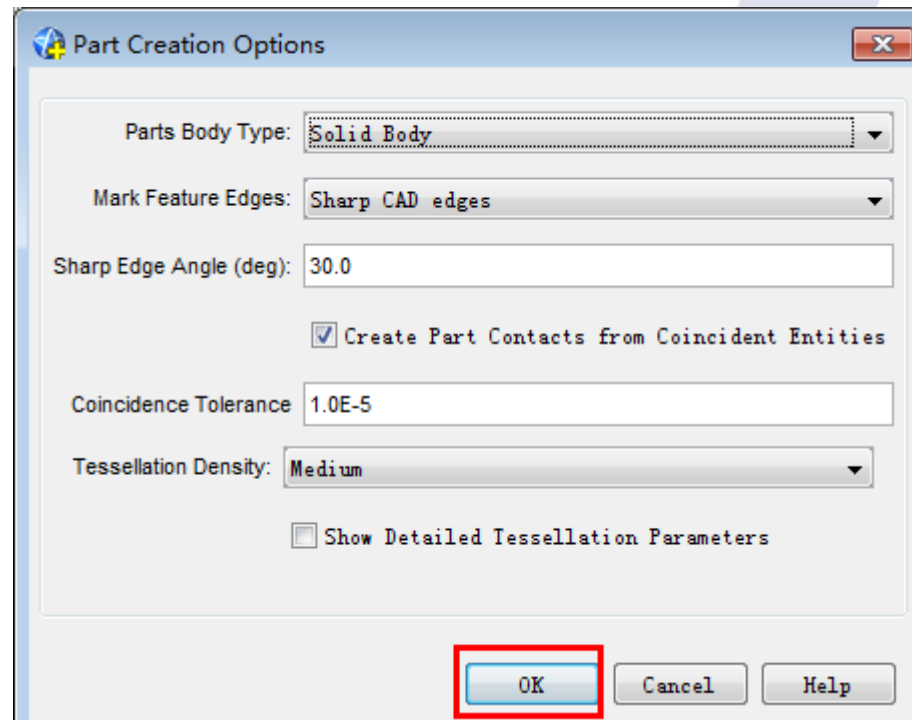
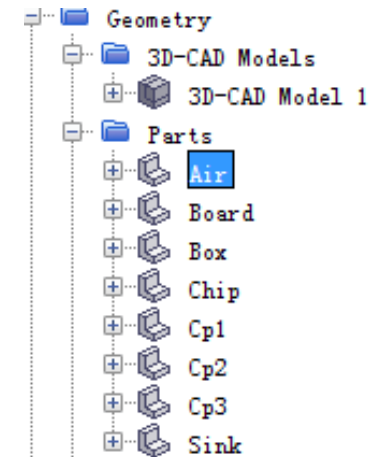
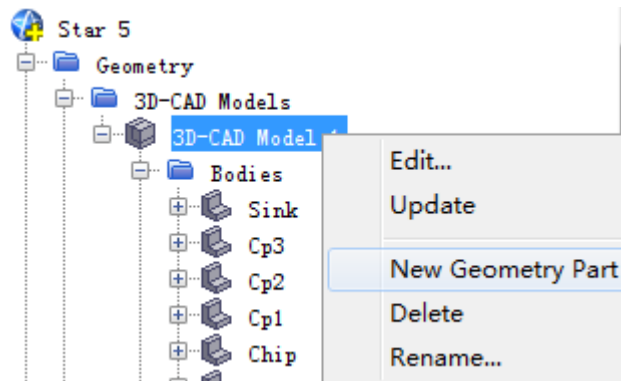
CHT Demo (表面重命名)

1. 将入口重命名为“Inlet”，出口命名为“Outlet”
2. 关闭3D-CAD。



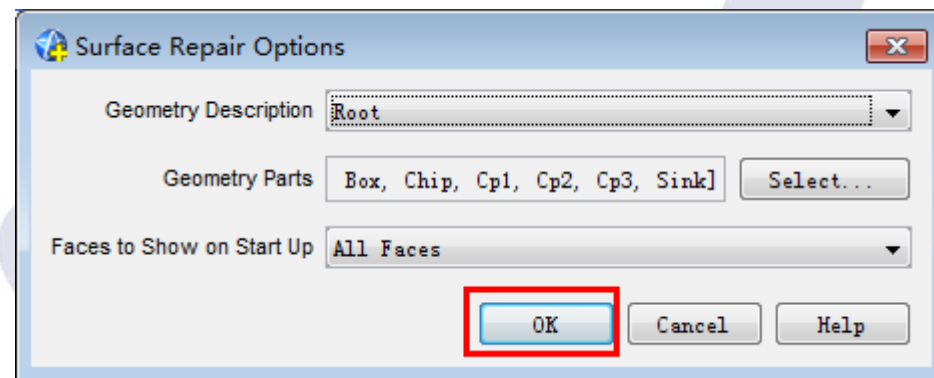
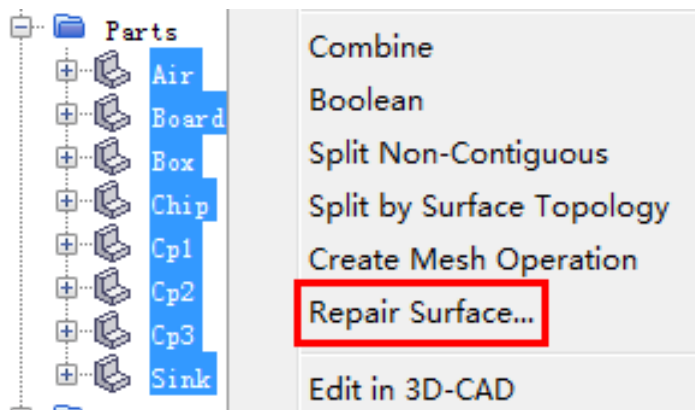
CHT Demo (生成Geometry Part)

1. 右击[3D-CAD Model 1] , 选择 [New Geometry Part]
 - 将会转换为Geometry Part。
2. 在“Part Creation Options”窗口中选择OK



CHT Demo (Surface Repair)

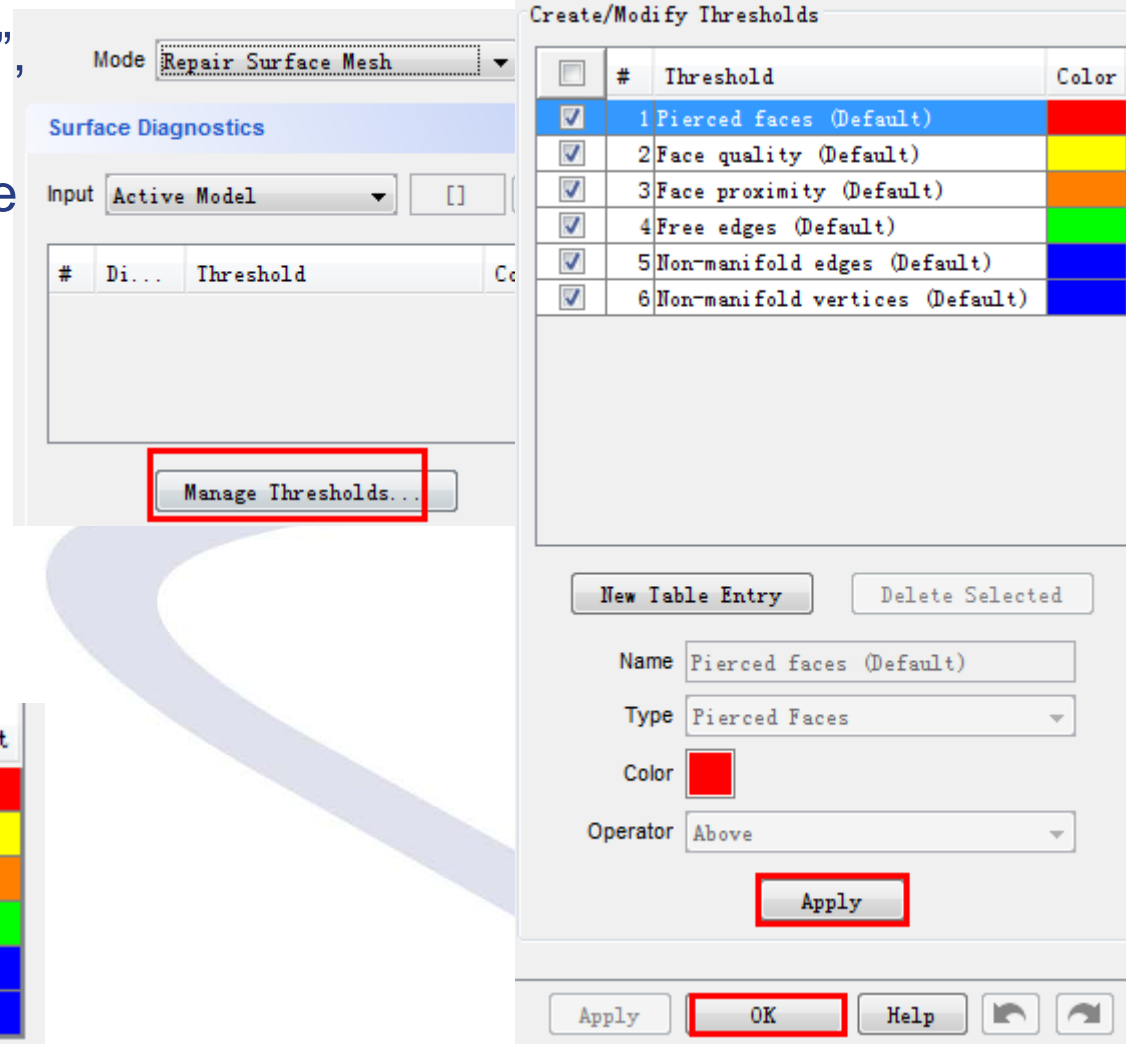
1. 选择所有的部件，右击，选择[Repair Surface]
 - 确认Part是否有问题。
2. 在Surface Repair Options窗口中选择OK。



CHT Demo (Surface Repair)

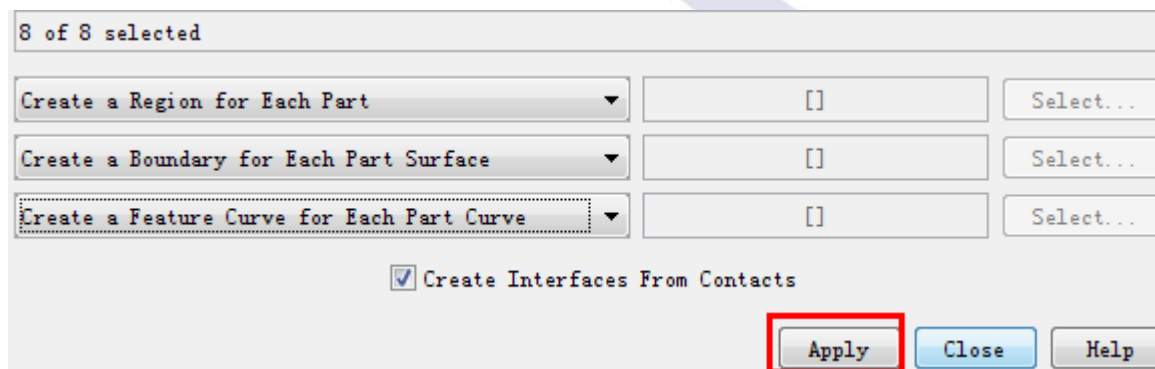
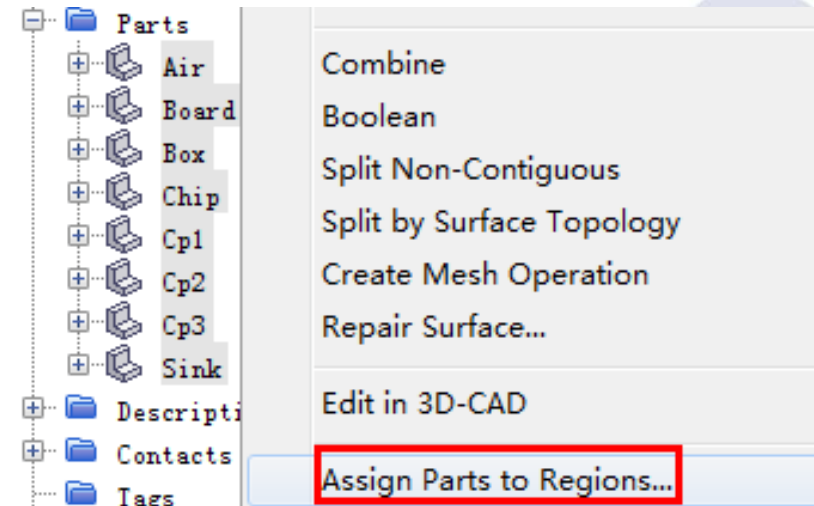
1. 选择“Manage Thresholds”, 点击[OK]
2. 存在“Face quality”和“Face proximity”的错误
 - 上述错误通过Surface Remesh会自动修复
3. 关闭该窗口。

#	Di...	Threshold	Count
1	☒	Pierced faces (Default)	0
2	☒	Face quality (Default)	512
3	☒	Face proximity (Default)	10
4	☒	Free edges (Default)	0
5	☒	Non-manifold edges (D...	0
6	☒	Non-manifold vertices...	0



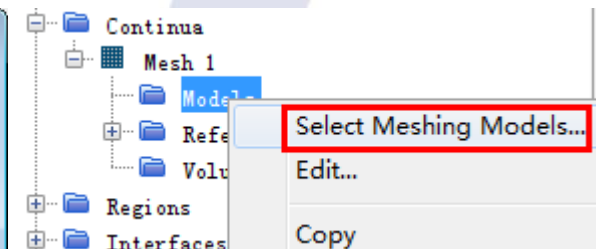
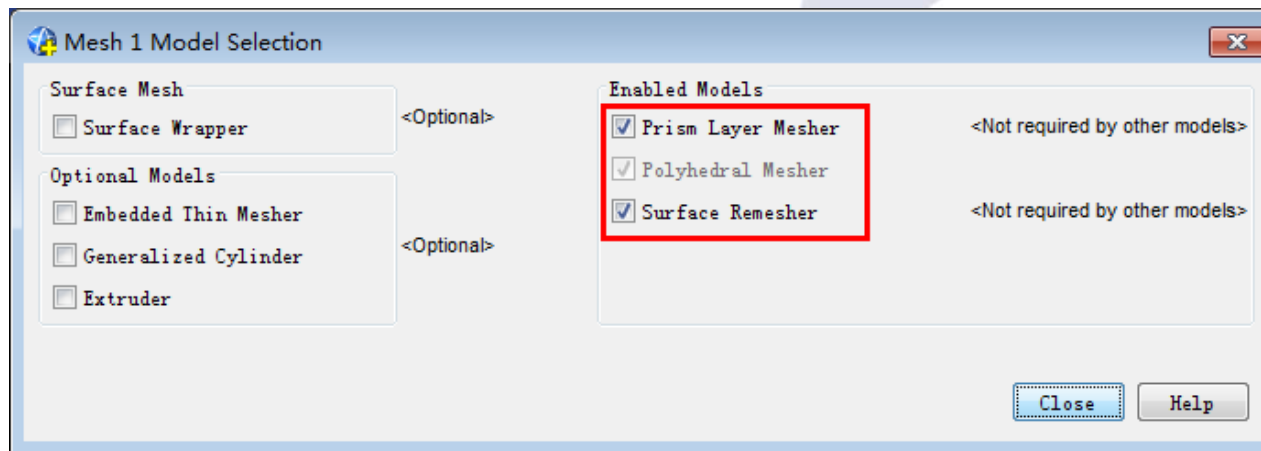
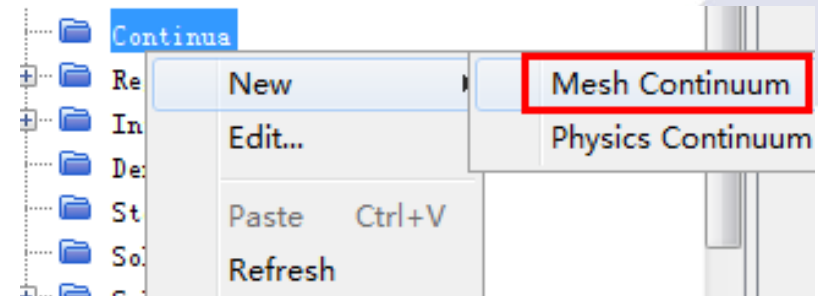
CHT Demo (从Part生成Region)

1. 选择所有的Part，右击，选择> [Assign Parts to Regions]
 - 为了生成体网格，必须从Part生成Region。
2. 在Assign Parts to Regions窗口中，分别选择：
 - Create One Region for All Parts
 - Create a Boundary for Each Part
 - Create One Feature Curve for All Part Curves



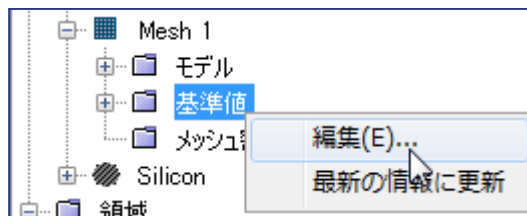
CHT Demo (定义Mesh Continua)

1. 右击[Continua] , 选择 [New]
> [Mesh Continua]。
 - 为了生成体网格, 必须先新建一个 Mesh Continua
2. 右击[Continua] > [Mesh 1] , 选择 [Select Meshing Models]
3. 选择Surface Remesher, Polyhedral Mesher和Prism Layer Mesher



CHT Demo (设定网格尺寸)

1. 右击[Continua] > [Mesh 1] > [Reference Values] , 选择[Edit]

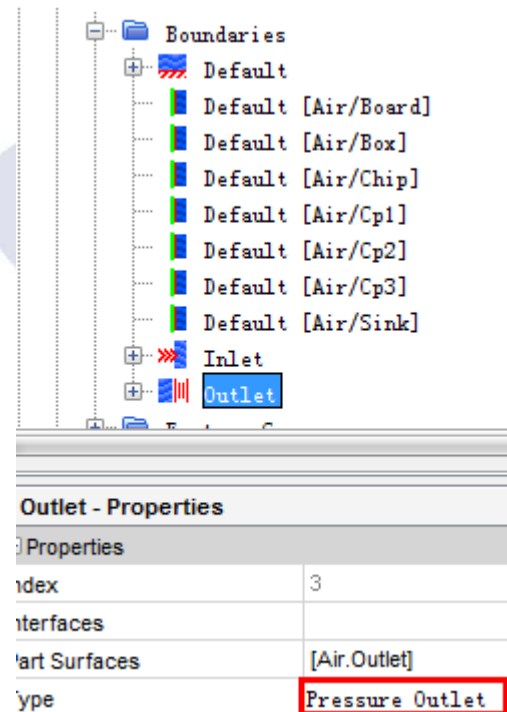
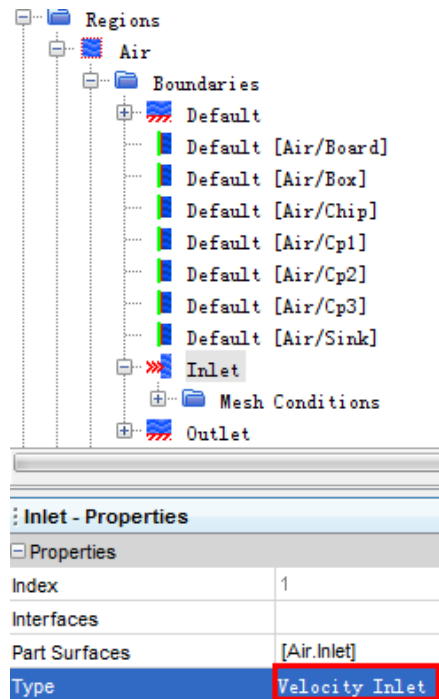


2. 根据右图进行设置。
 - [Base Size]设置为5mm
 - [Number of Prism Layers]设为2
 - [Prism Layer Thickness]设为2.5mm。因为Prism Layer Stretching设为为1.1 , 一层的厚度近似约为1.25mm。

[-] Reference Values	
[-] Base Size	
Value	5.0 mm
[+] Automatic Surface Repair	
[+] CAD Projection	
[-] Number of Prism Layers	
Number of Prism Layers	2
[-] Prism Layer Stretching	
Prism Layer Stretching	1.1
[-] Prism Layer Thickness	
Size Type	Absolute
[-] Absolute Size	
Value	2.5 mm

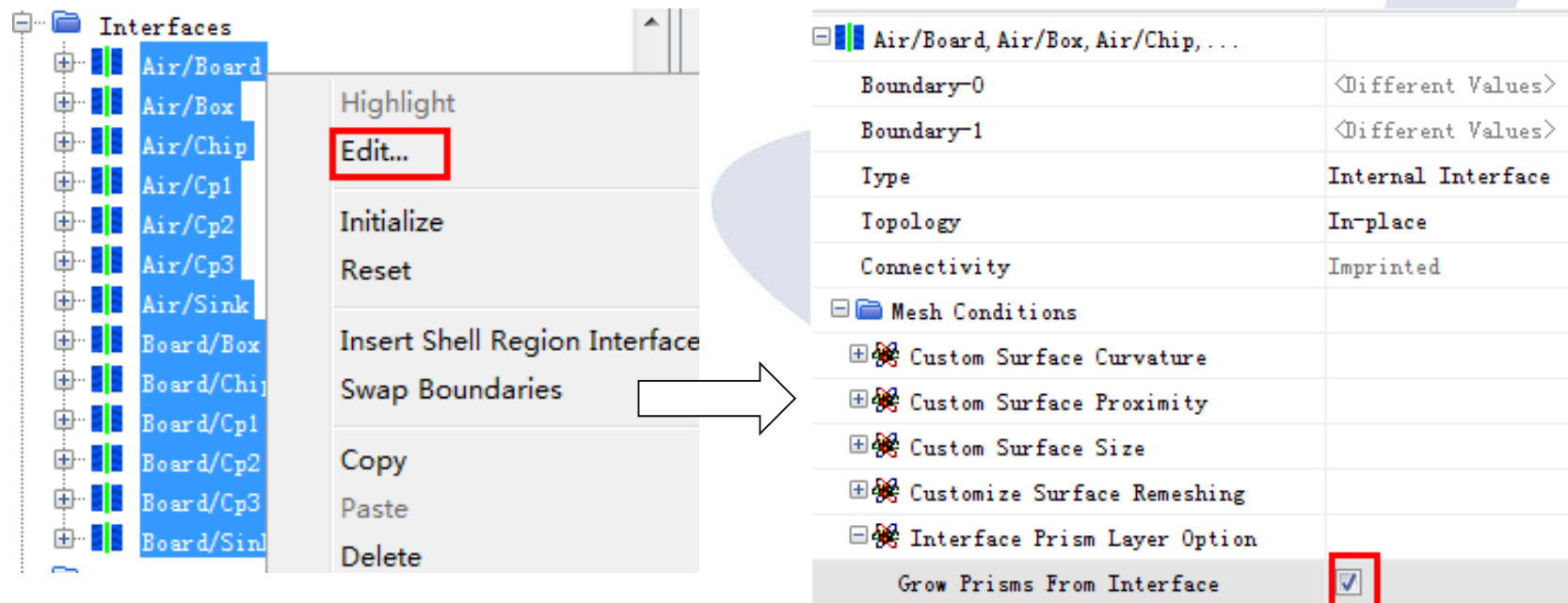
CHT Demo (改变边界类型)

1. [Region] > [Air] > [Boundaried] > [Inlet] ,在Properties中将其类型改为Velocity Inlet
 2. [Region] > [Air] > [Boundaries] > [Outlet] ,在Properties中将其类型改为Pressure Outlet
- 为了不在Inlet以及outlet处生成边界层，需要在生成网格之前提前指定其Type



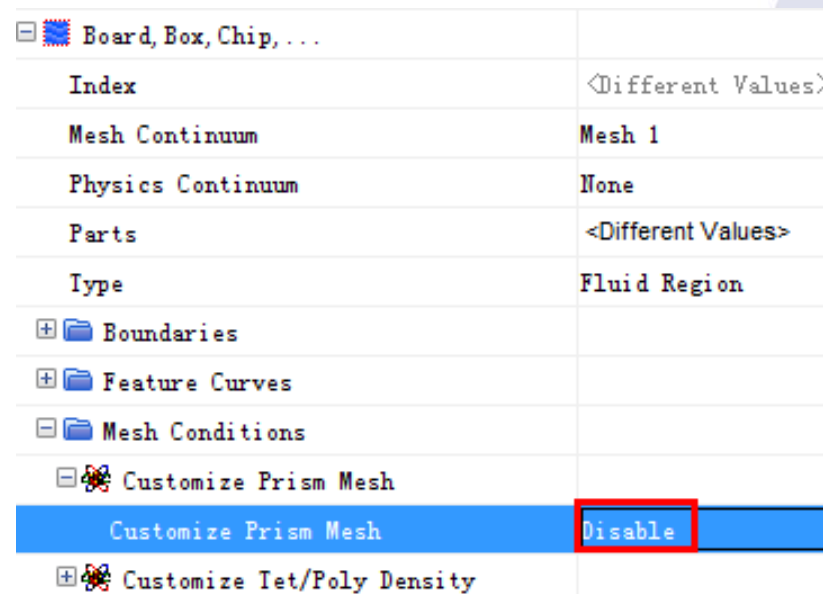
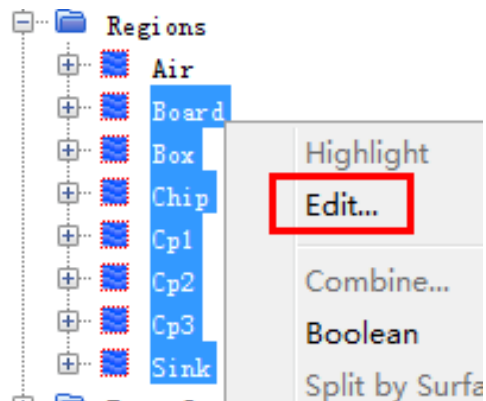
CHT Demo (interface上的边界层设置)

1. 选择所有的interface，右击，选择[Edit]
 - 需要在interface上生成边界层。
2. 勾选[Mesh Conditions] > [Interface Prism Layer Option] > [Grow Prisms From Interface]



CHT Demo (取消固体部分边界层的生成)

1. 选择所有的固体Region，右击，选择[Edit]
2. [Mesh Conditions] > [Customize Prism Mesh]，选择”Disable”
 - 固体侧将不再生成边界层。

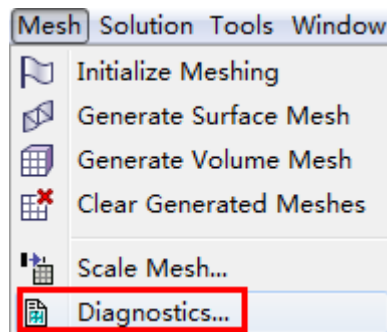


CHT Demo (生成网格)

1. 点击Generate Volume Mesh生成体网格。



1. 选择[Mesh] > [Diagnostic]



- Face Validity达到1.00，可以认为网格质量良好

--- Overall Statistics:

-> ENTITY COUNT:

Cells: 115601

Faces: 622368

Verts: 519928

-> MESH VALIDITY:

Mesh is topologically valid and has no negative volume cells.

-> FACE VALIDITY STATISTICS:

Minimum Face Validity: 1.000000e+00

Maximum Face Validity: 1.000000e+00

Face Validity < 0.50	0	0.000%
0.50 <= Face Validity < 0.60	0	0.000%
0.60 <= Face Validity < 0.70	0	0.000%
0.70 <= Face Validity < 0.80	0	0.000%
0.80 <= Face Validity < 0.90	0	0.000%
0.90 <= Face Validity < 0.95	0	0.000%
0.95 <= Face Validity < 1.00	0	0.000%
1.00 <= Face Validity	115601	100.000%

-> VOLUME CHANGE STATISTICS:

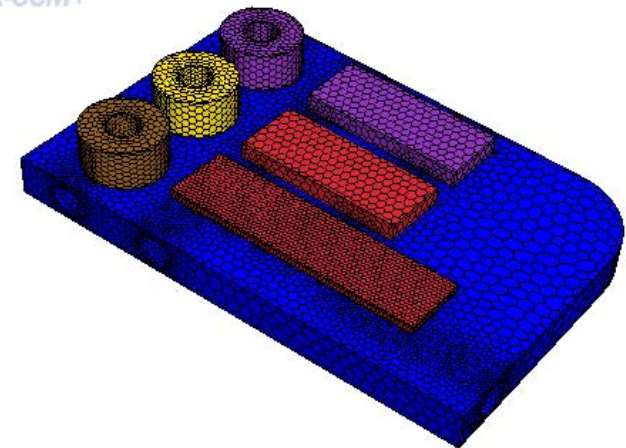
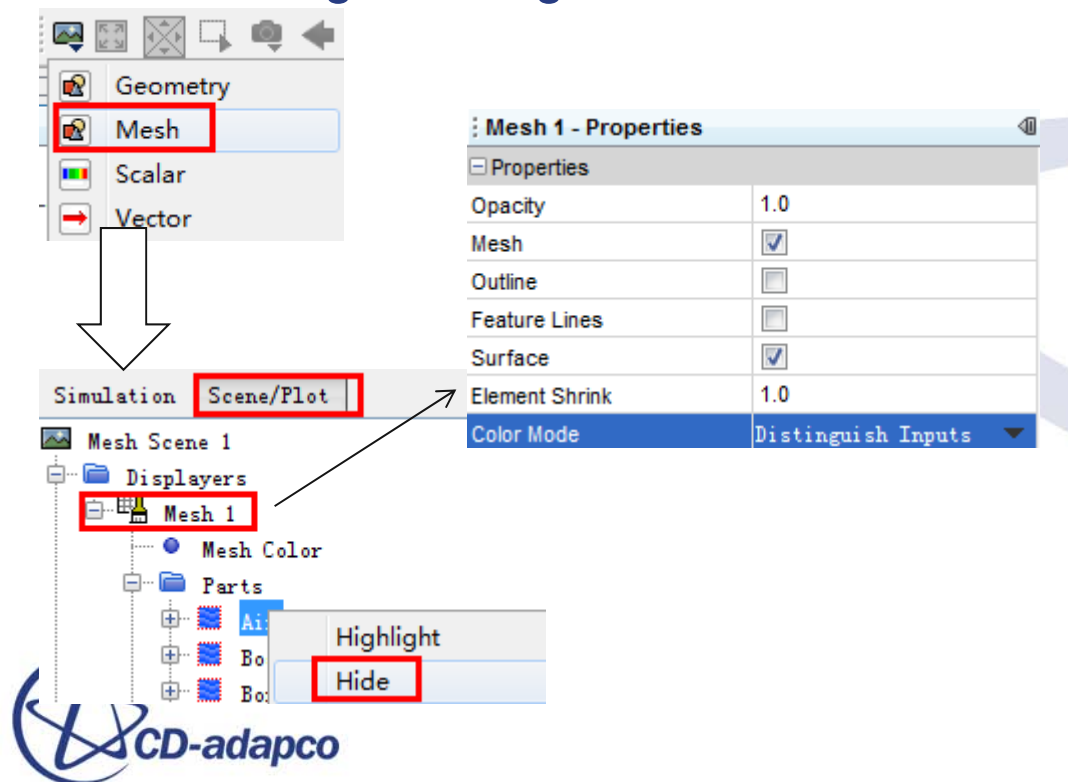
Minimum Volume Change: 1.014788e-03

Maximum Volume Change: 1.000000e+00

Volume Change < 0.000000e+00	0	0.000%
0.000000e+00 <= Volume Change < 1.000000e-06	0	0.000%
1.000000e-06 <= Volume Change < 1.000000e-05	0	0.000%
1.000000e-05 <= Volume Change < 1.000000e-04	0	0.000%
1.000000e-04 <= Volume Change < 1.000000e-03	0	0.000%
1.000000e-03 <= Volume Change < 1.000000e-02	111	0.096%
1.000000e-02 <= Volume Change < 1.000000e-01	4538	3.926%
1.000000e-01 <= Volume Change <= 1.000000e+00	110952	95.978%

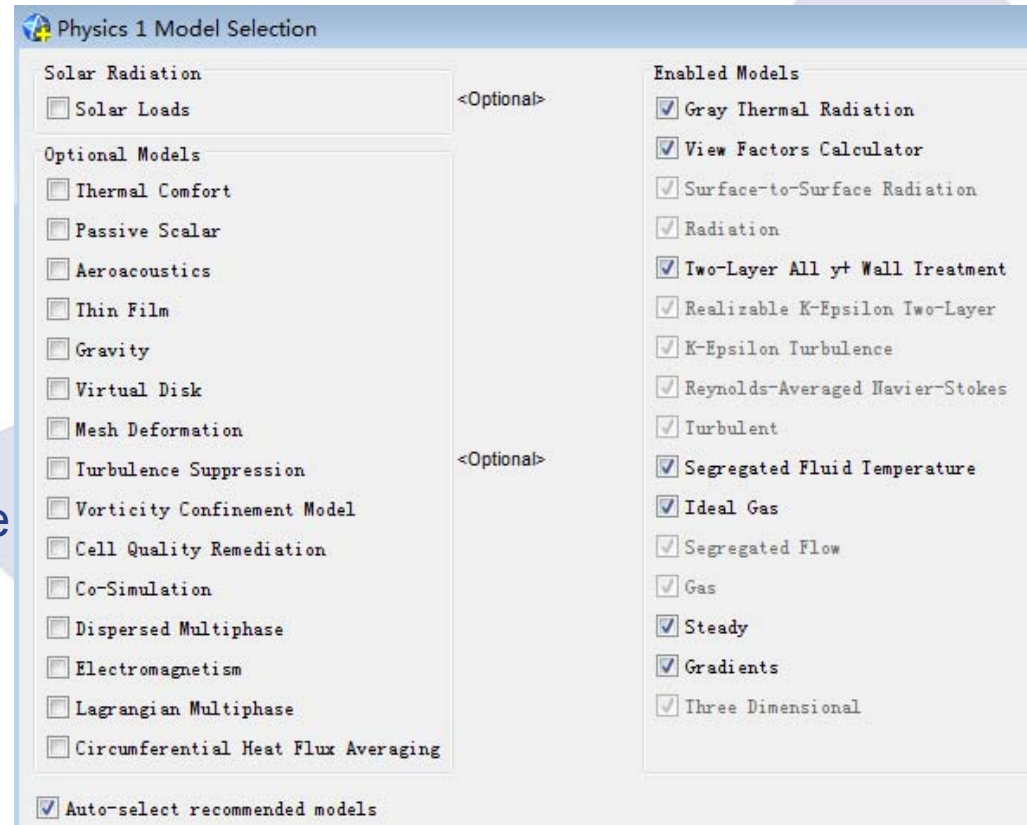
CHT Demo (检查生成的网格)

1. 打开Mesh Scene 1
2. 打开[Scene/Plot]标签，右击[Mesh Scene 1] > [Parts] > [Air],选择“Hide”
3. 在[Mesh Scene 1]的Property中，选择 [Color Mode],设置为“Distinguish Regions”



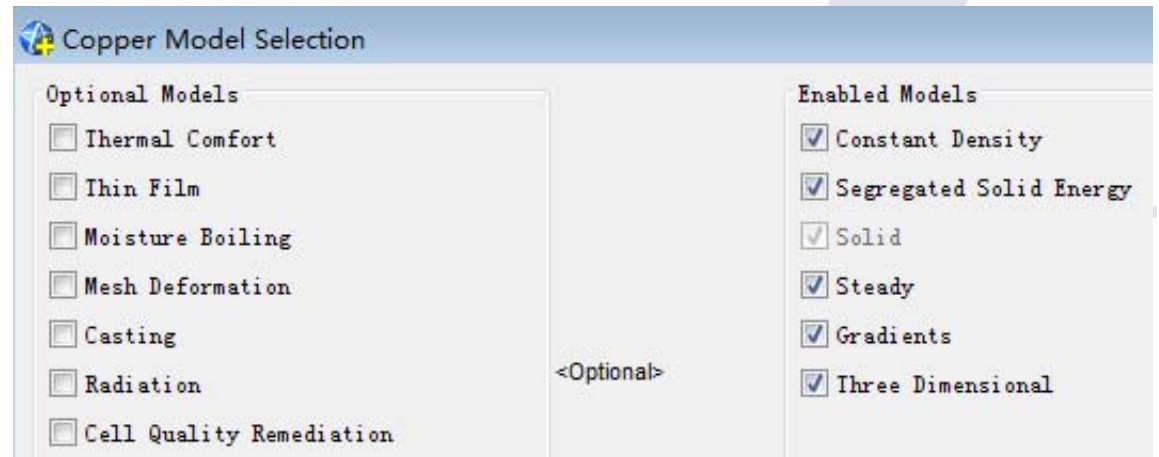
CHT Demo (设定物理连续体)

1. 将[Continua] > [Physics 1] 重命名为 [Air]。
2. 右击[Continua] > [Air] , 选择 [Select Models]
 - Steady
 - Gas
 - Ideal Gas
 - Turbulent
 - Segregated Fluid Temperature
 - Radiation
 - Gray Thermal Radiation
 - Surface-to-Surface Radiation



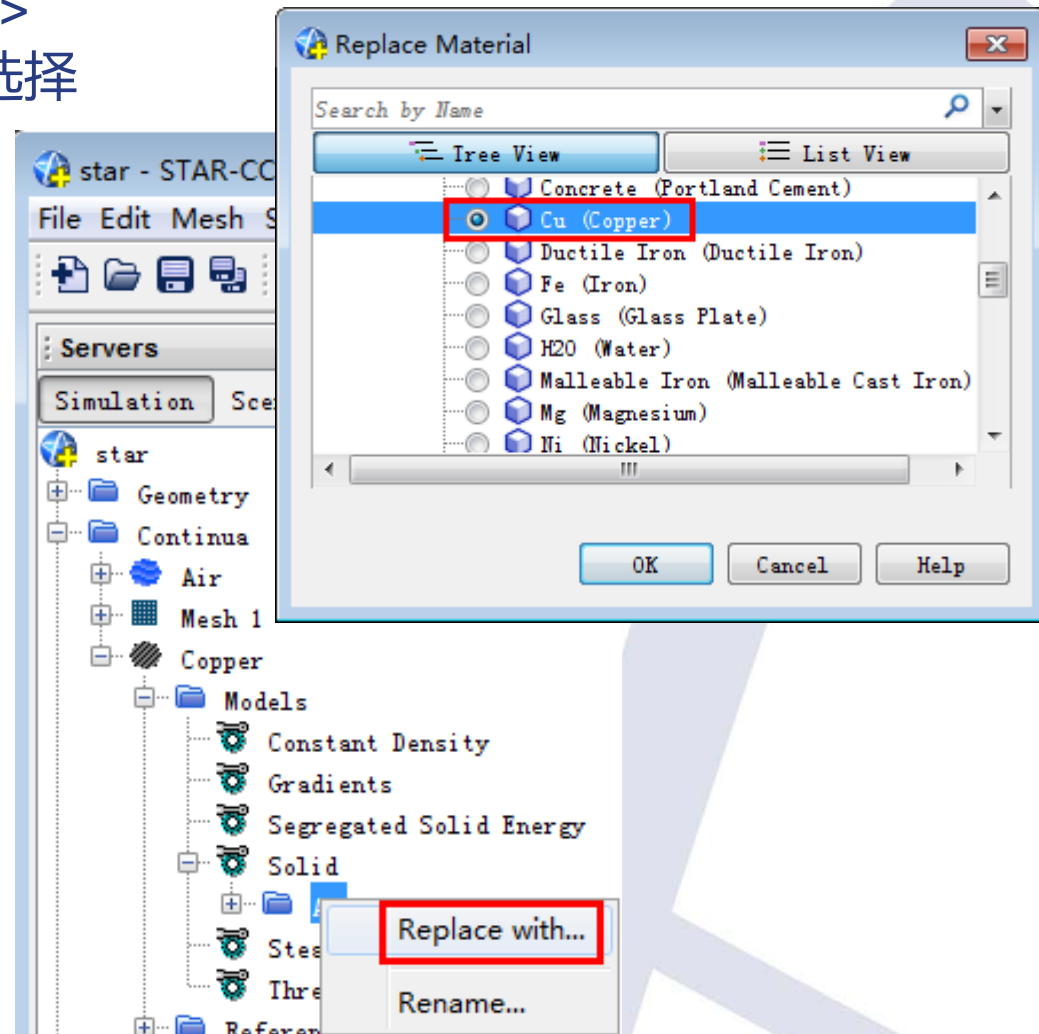
CHT Demo (设定Copper的物理连续体)

1. 右击[Continua] > [New] > [Physical Continua]
2. 将新生成的物理连续体命名为“Copper”
3. 选择以下的物理模型
 - Steady
 - Solid
 - Constant Density
 - Segrated Solid Energy



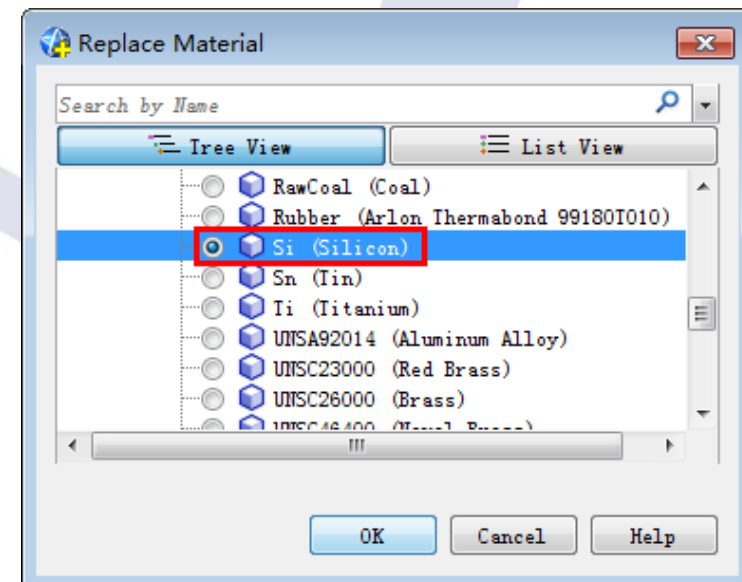
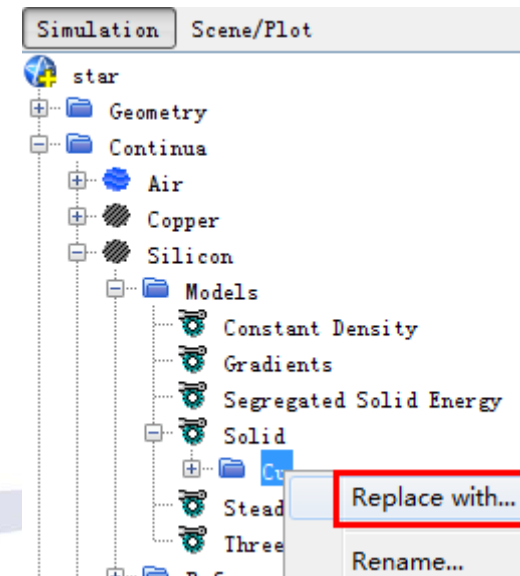
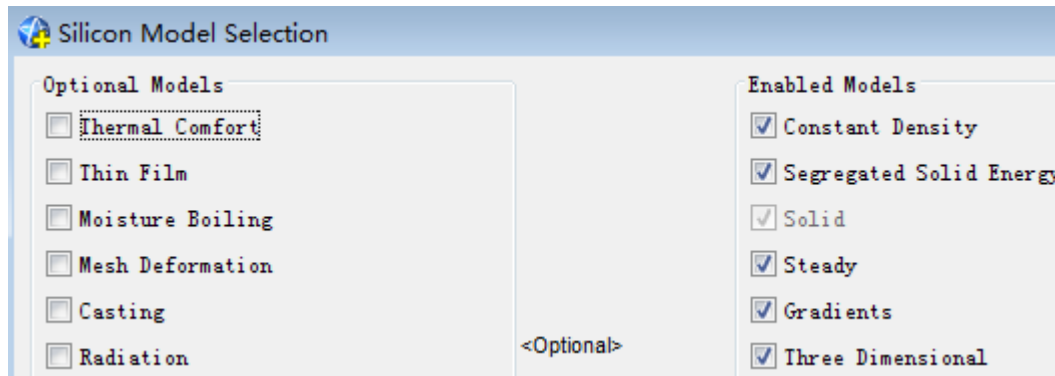
CHT Demo (设定Copper的物理连续体)

1. 右击[Continua] > [Copper] > [Models] > [Solid] > [Al] , 选择 [Replace with]
2. 选择[Material Database] > [Standard] > [Solids] > Cu (Copper)



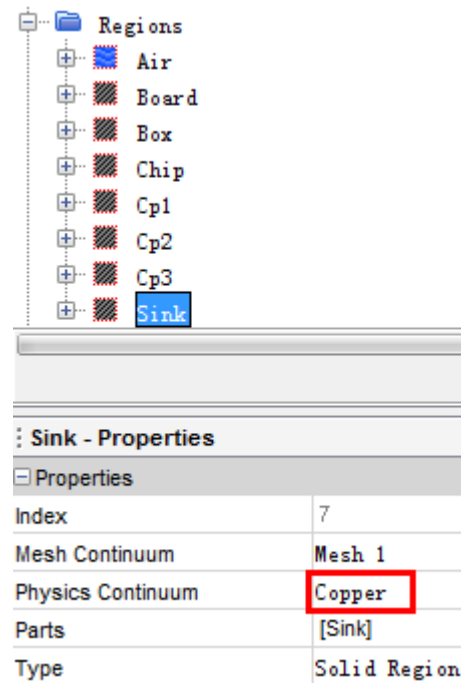
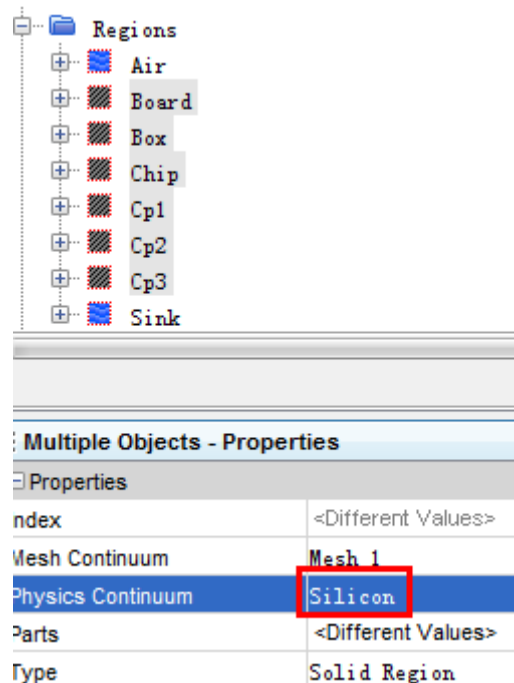
CHT Demo (设定Silicon的物理连续体)

1. Silicon的物理连续体生成的方法与Copper相同，所不同的就是在Material中选择了[Si(Silicon)]



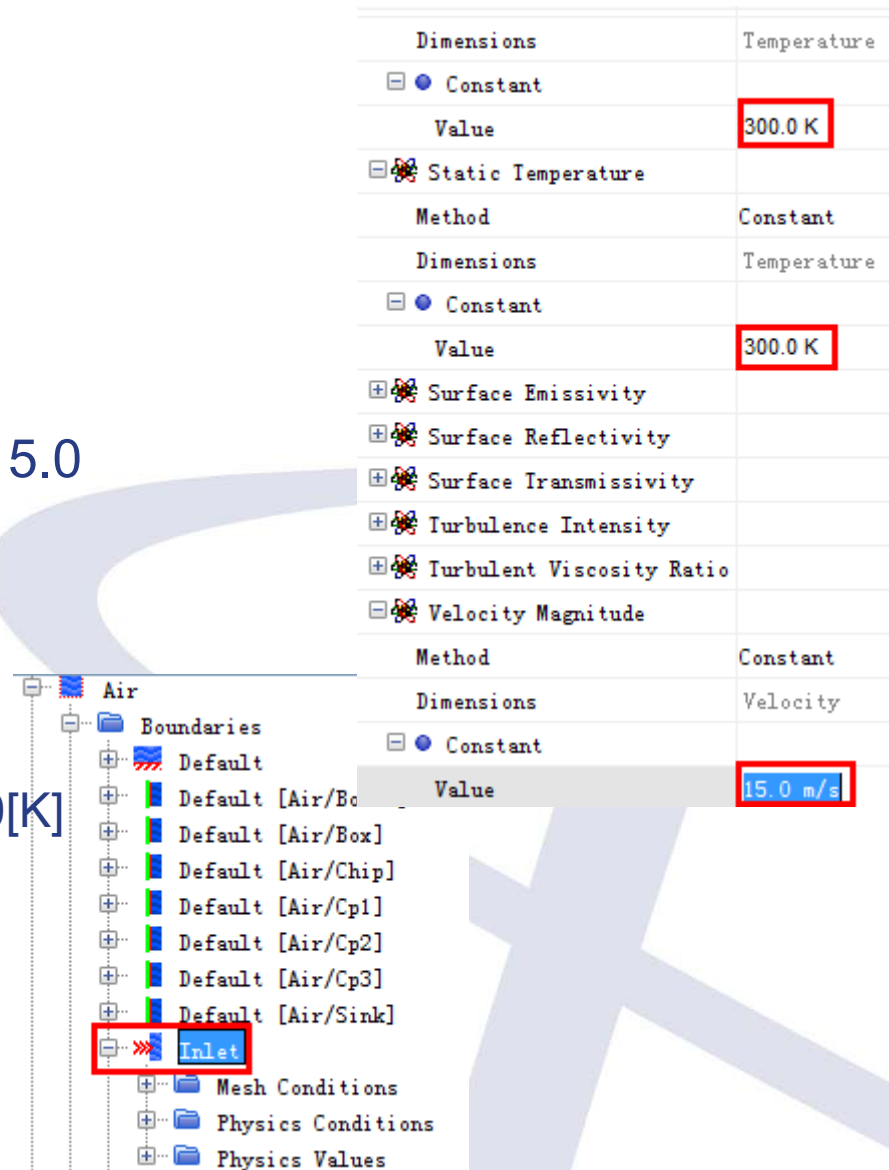
CHT Demo (改变Region的物理连续体)

1. 将除了[Air]和[Sink]以外的Region的物理域设置为”Silicon”，将[Sink]设置为“Copper”



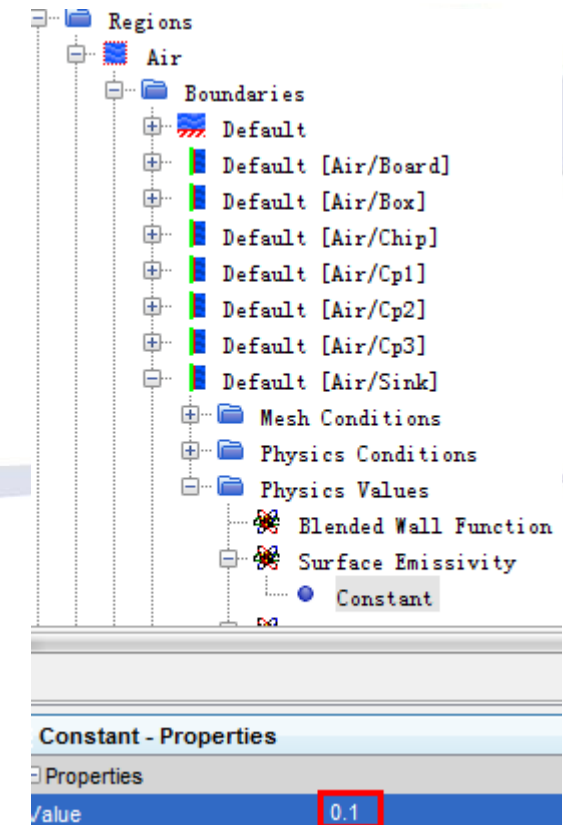
CHT Demo (Inlet & Outlet的边界条件)

1. 右击[Region] > [Air] > [Boundaries] > [Inlet] , 选择 [Edit]
2. [Radiation Temperature]和 [Static Temperature]设置为 300K。
3. [Velocity Magnitude]设置为15.0 [m/s]
4. Outlet保持默认的边界条件
 - Pressure : 0[Pa]
 - Static Temperature : 300[K]
 - Radiation Temperature : 300[K]



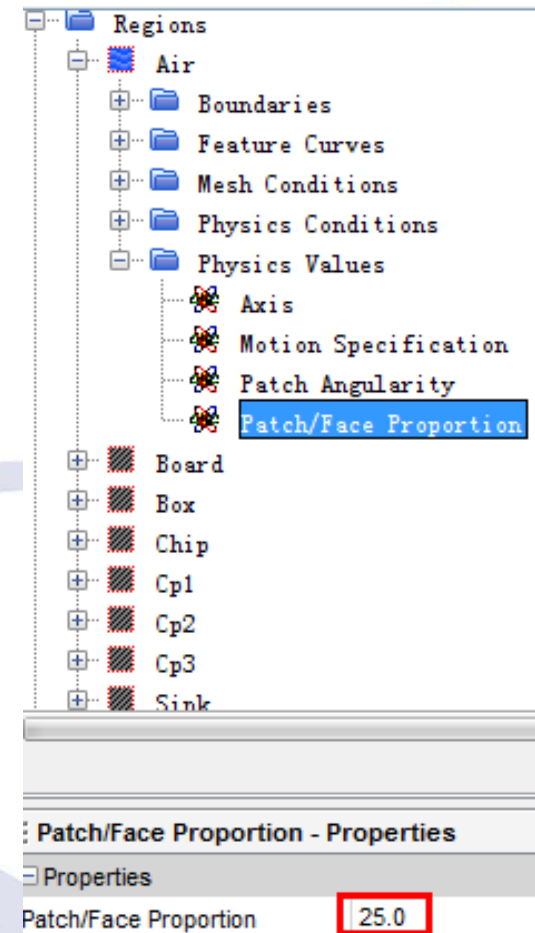
CHT Demo (指定表面发射率)

1. [Region] > [Air] > [Boundaries] > [Default (Air/Sink)] > [Physics Values] > [Surface Emissivity] > [constant] , 设置其值为0.1
 - Silicon的表面发射率默认为0.8
 - 因为固体都是不能穿透的, 所以只在流体域内设置辐射的相关物性



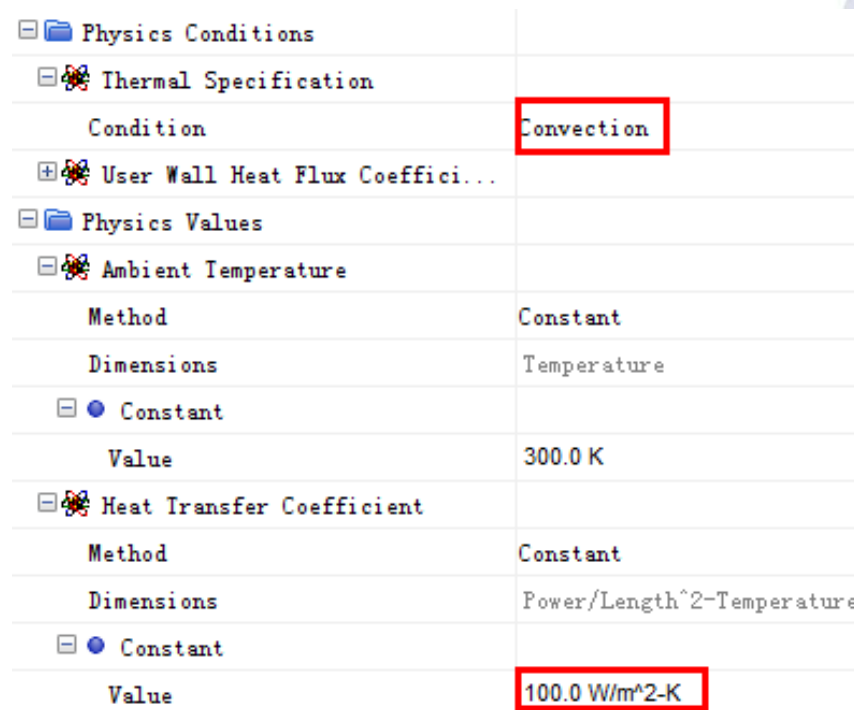
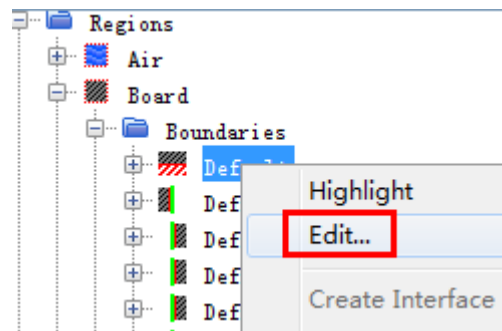
CHT Demo (设定Patch/Face Proportion)

1. [Region] > [Air] > [Physics Values] > [Patch/Face Proportion], 设置其值为25%
 - 4个网格的面组成一个Patch
 - 在Patch/Face Proportion属性窗口中可以调整Patch数



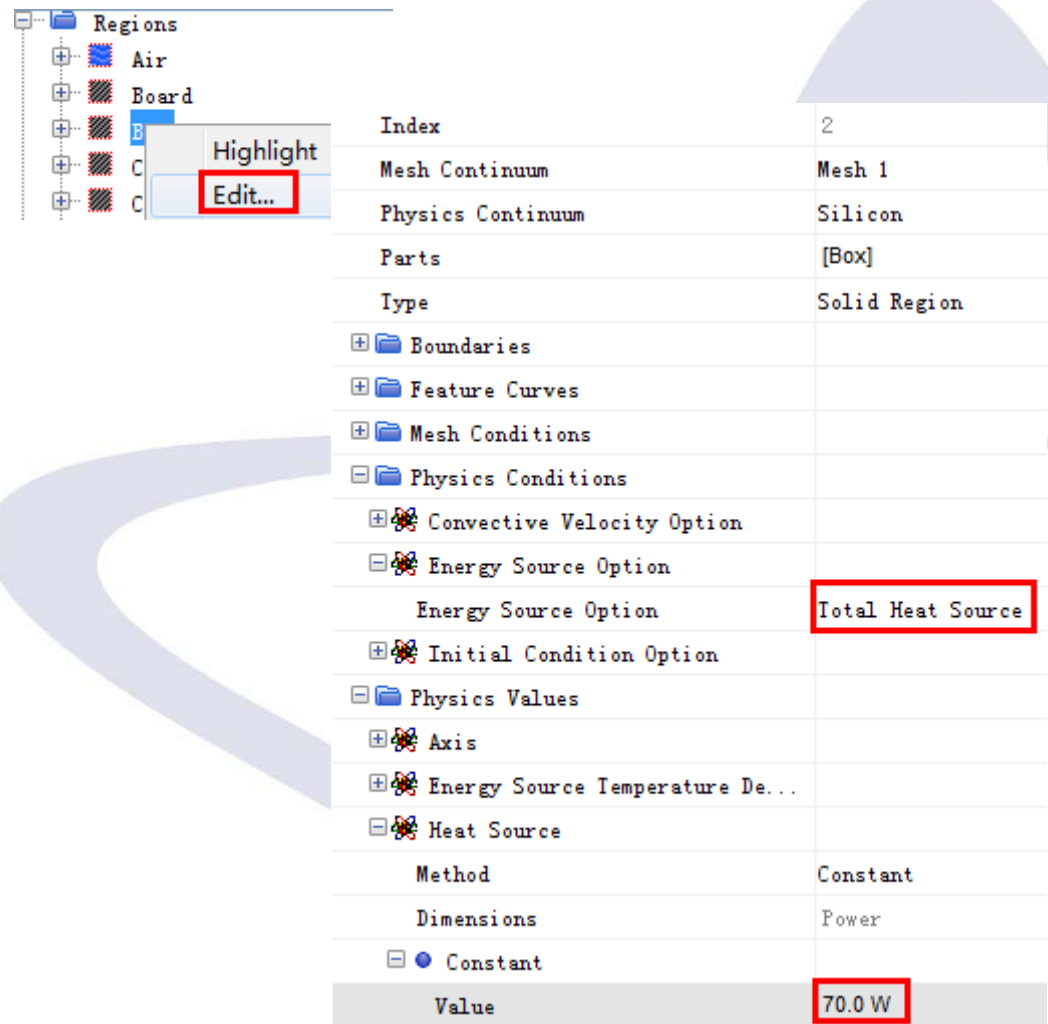
CHT Demo (Board热边界条件)

1. 右击[Region] > [Board] > [Boundaries] > [Default] , 选择[Edit]
2. [Physics Conditions] > [Thermal Specification],选择“Convection”
3. [Physics Values] > [Heat Transfer Coefficient] , 设置其值为 100[W/m²-K]



CHT Demo (Box Heat Source设定)

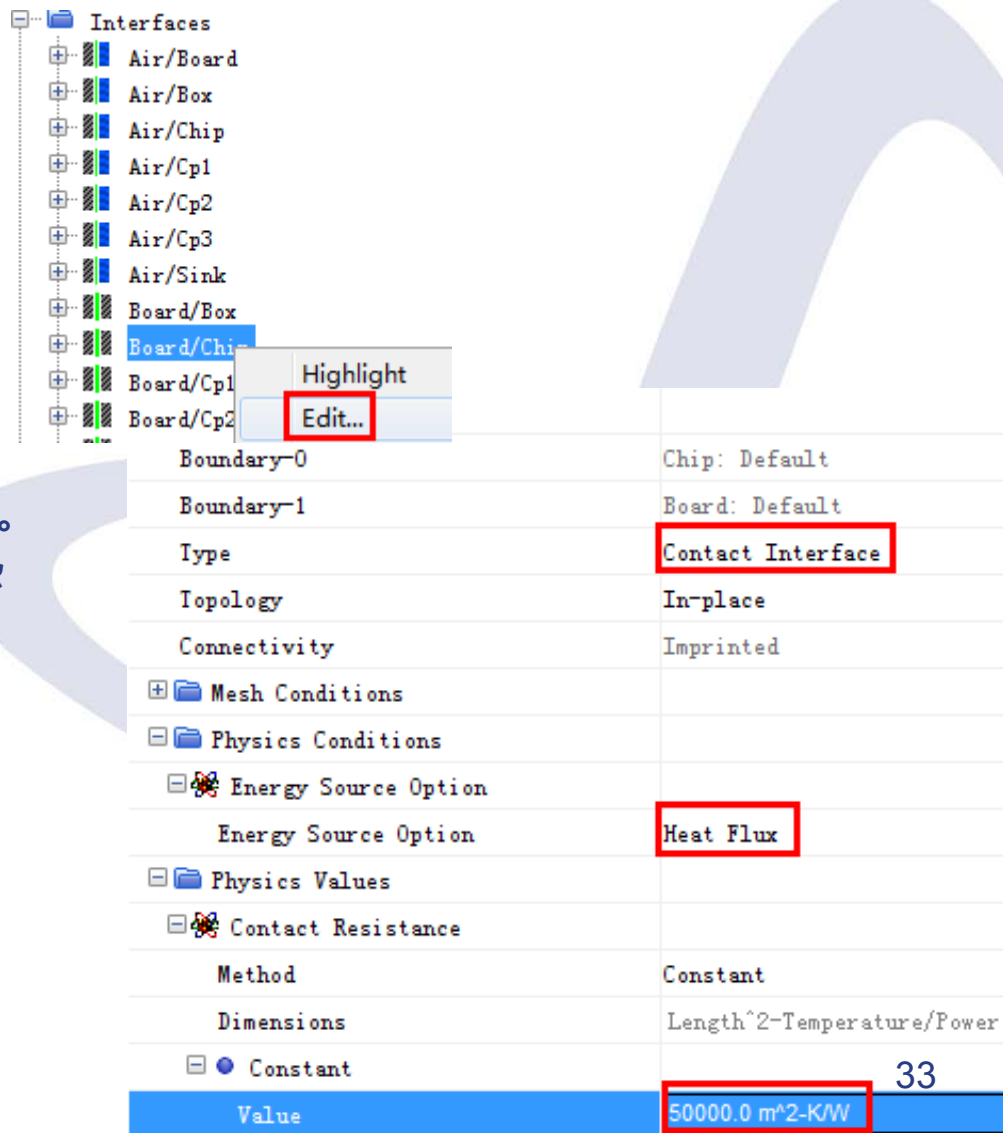
1. [Region] > [Box] > [Physics Condition] > [Energy Source Option], 选择[Total Heat Source]选项
2. 右击[Physics Values], 选择[Edit] > [Heat Source] > [Constant], 设定其值为70[W]



Index	
2	
Mesh Continuum	Mesh 1
Physics Continuum	Silicon
Parts	[Box]
Type	Solid Region
+ Boundaries	
+ Feature Curves	
+ Mesh Conditions	
- Physics Conditions	
+ Convective Velocity Option	
- Energy Source Option	
Energy Source Option	Total Heat Source
+ Initial Condition Option	
- Physics Values	
+ Axis	
+ Energy Source Temperature De...	
- Heat Source	
Method	Constant
Dimensions	Power
- Constant	
Value	70.0 W

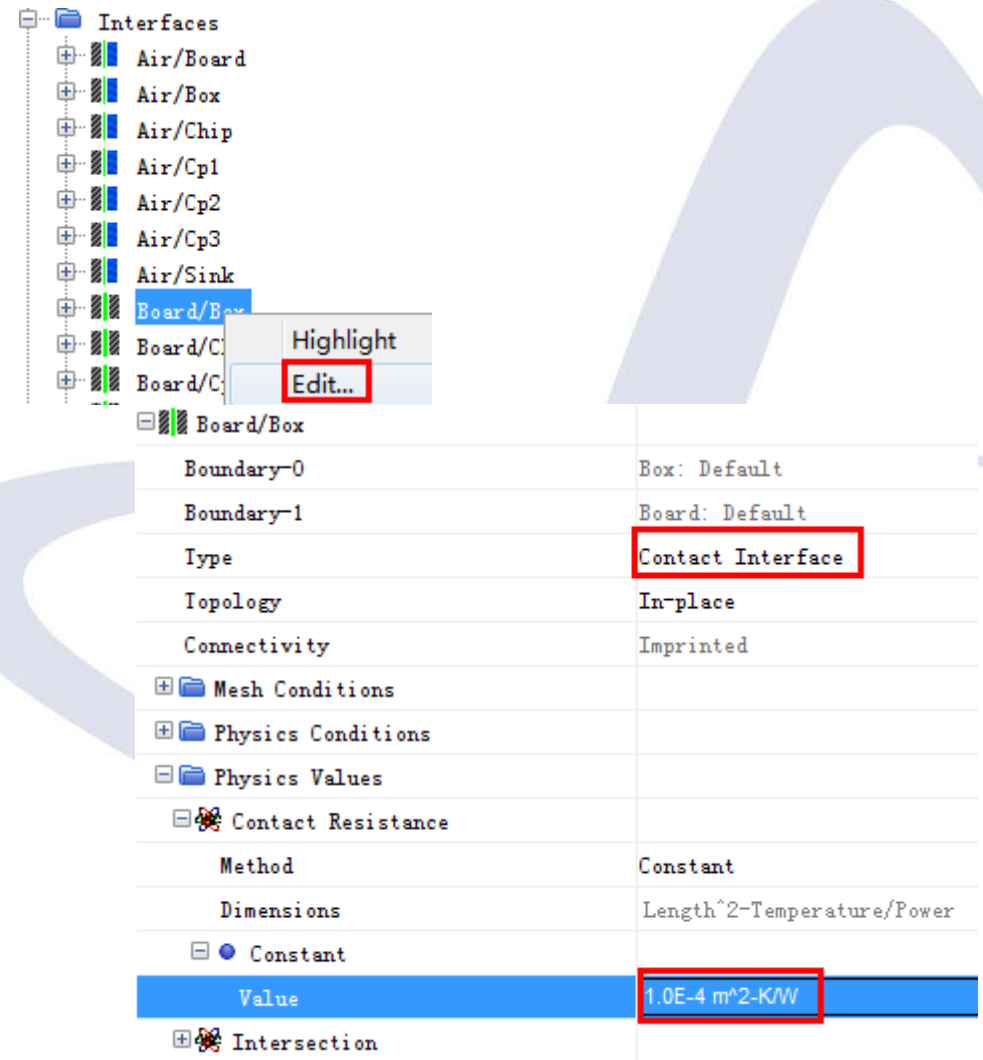
CHT Demo (Board-Chip间发热设定)

1. 右击[Interface] > [Board/Chip] 选择[Edit]。
2. 将[Type]改为“Contact Interface”
3. 选择[Physics Values] > [Heat Flux] > [Constant],指定其值 50000[W/m²].
 - 从interface到各region都有放热。根据各region的热阻值决定放热的比例。



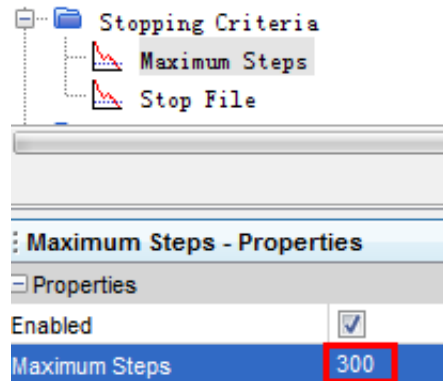
CHT Demo (Box-Board间接触热阻设定)

1. 选择[Interface] > [Board/Box] > [Physics Values] > [Contact Resistance] > [Constant]
 - 定义Board/Box间的接触热阻
2. 设置其值为 $1e-4$ [m^2-K/W]

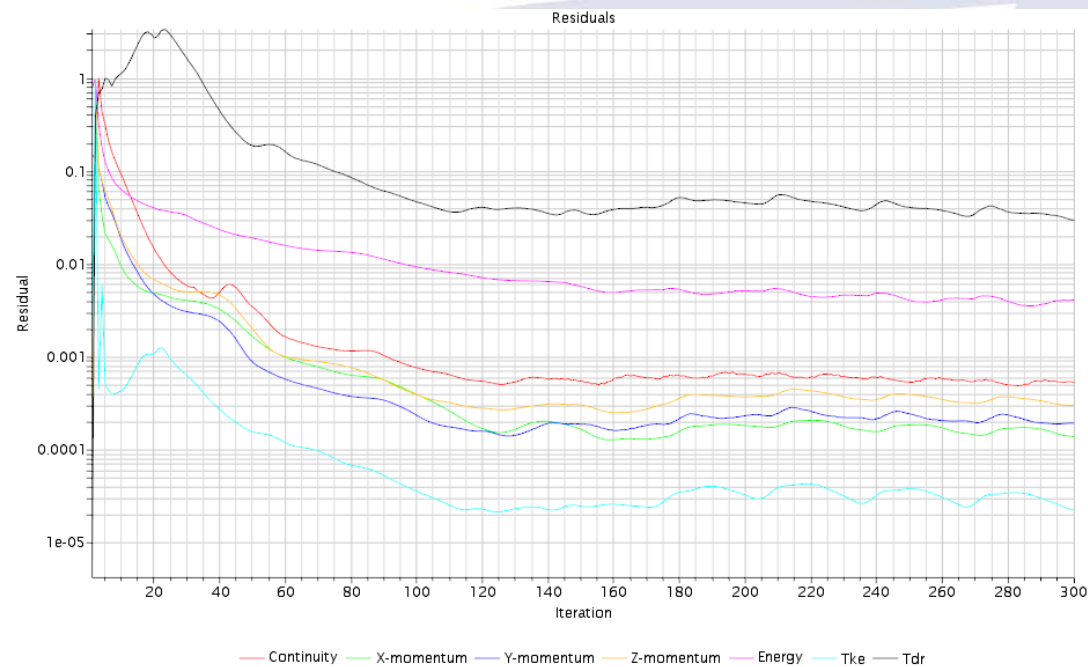


设定最大迭代步数& 实行计算

1. 在[Stopping Criteria] > [Maximum Steps] , 设置其值为300。

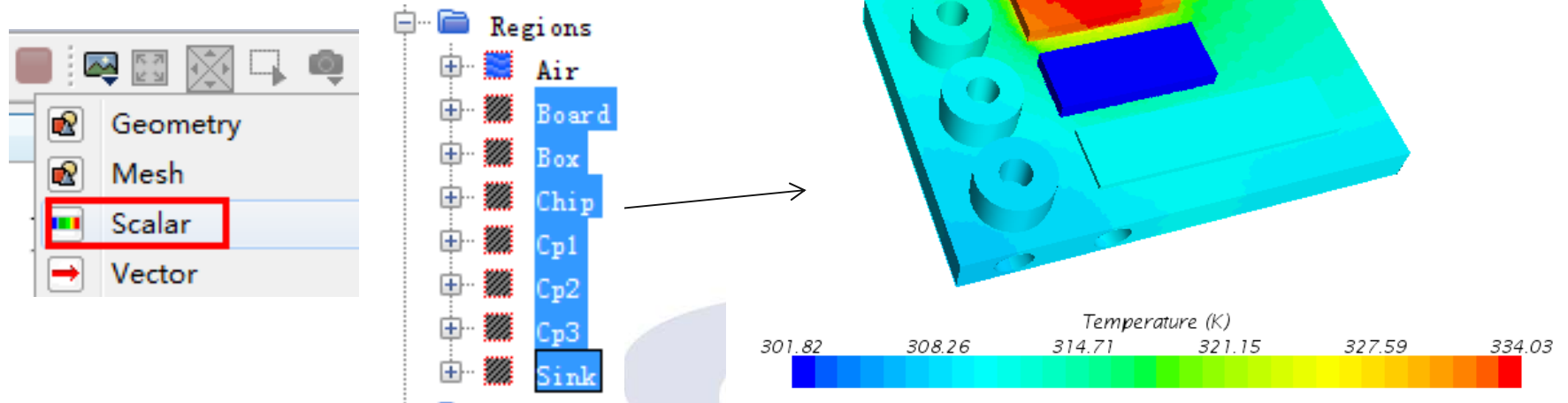


2. 开始计算。



温度分布

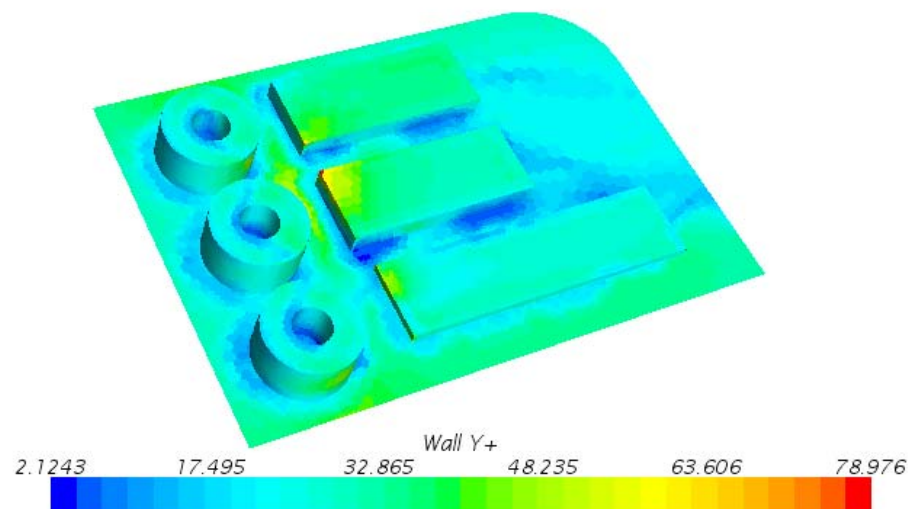
1. 新建一个标量云图，将固体域拖到该标量云图中。



2. 右击Select Function，选择Temperature

壁面的 y^+

1. 新建标量云图，鼠标拖动流体域侧的interface到创建的云图中。
2. 右击Select Function，选择Wall Y^+
 - 通过high- y^+ 近壁面处理（壁面函数）时，需要 y^+ 值在适当范围内。
 - 图中可以观察到，实际计算中的 y^+ 比 y^+ 的目标值偏低。
 - y^+ 偏低的区域基本上位于流体的壁面分离区域。
 - 壁面函数是流体的流动充分发展（理想状态）情况下进行定义，在此范围以外区域并不需要保证 y^+ 在对应的取值范围内。



Best Practice: 传热系数

- 对流传热系数通过以下的公式进行定义。

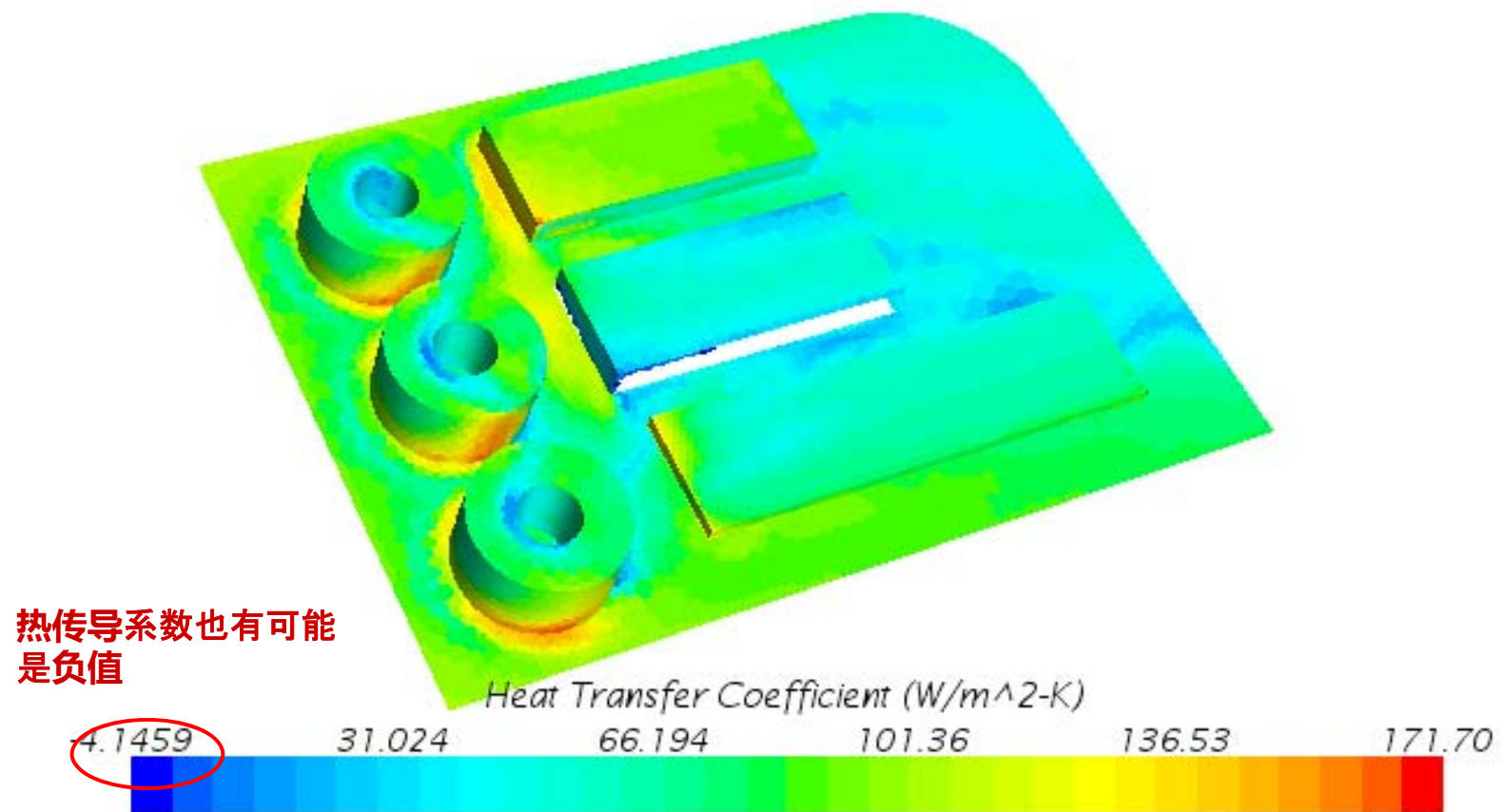
$$h = \frac{q''_{wall}}{(T_{wall} - T_{fluid})}$$

- 流体温度选择不一样。HTC的计算结果也不一样。

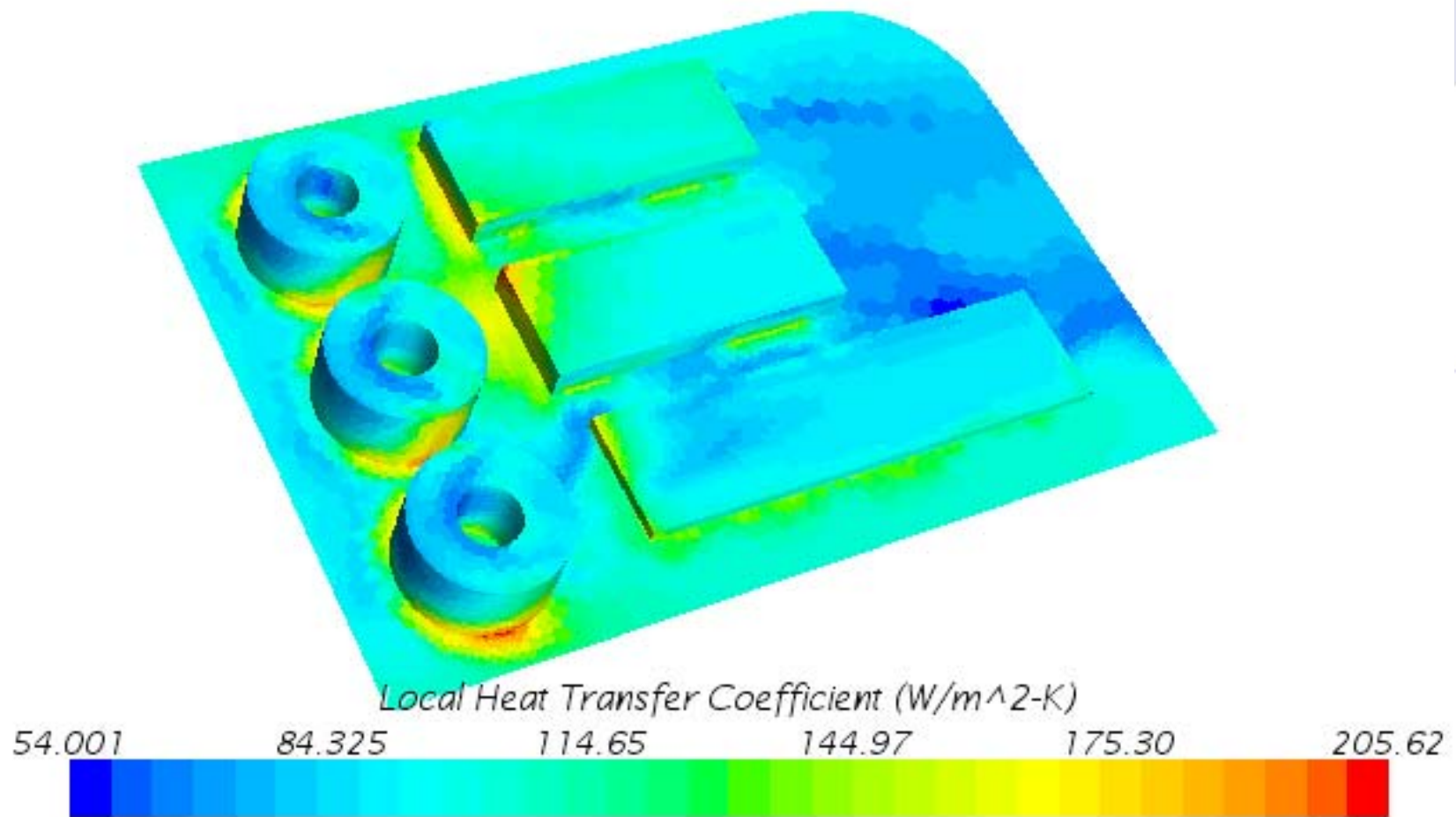
Best Practice STAR-CCM+中的传热系数

- Heat Transfer Coefficient:
 - » 计算中使用到了壁面温度以及热流量、流体温度另外指定。
 - 因为使用了用户指定的温度，所以不再考虑流体的温度分布特性
- Local Heat Transfer Coefficient:
 - » 使用近壁面处的流体温度计算传热系数。
 - 选取近壁面第一层网格的流体温度。
 - 通过该方法计算得到的传热系数与网格尺寸有关。
- Specified y+ Heat Transfer Coefficient:
 - » 使用指定的y+位置处的流体温度值。
 - 不再考虑流体的温度分布。
 - 排除了壁面第一层网格的影响
 - » 推荐使用“Heat Transfer Coefficient”和“Local Heat Transfer Coefficient”一起进行判断。

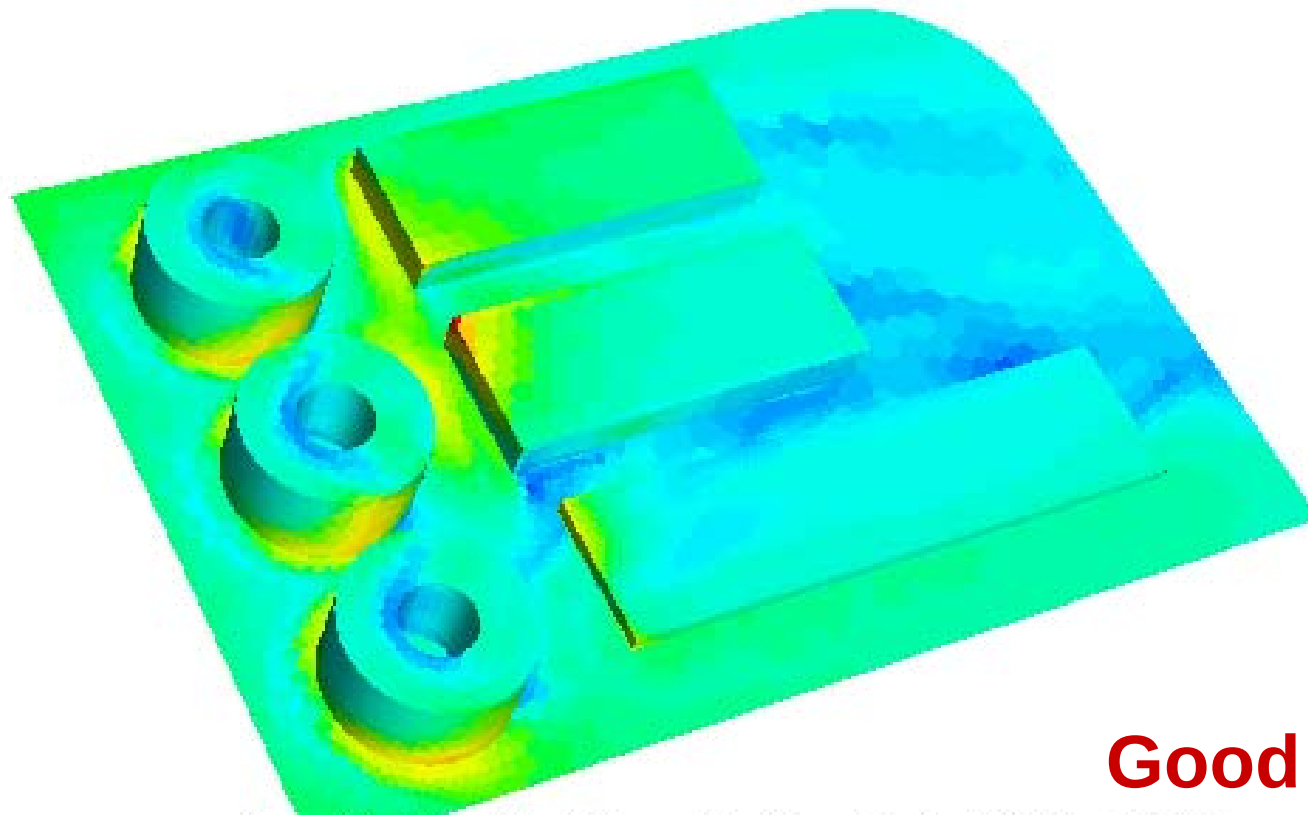
Heat Transfer Coefficient



Local HTC

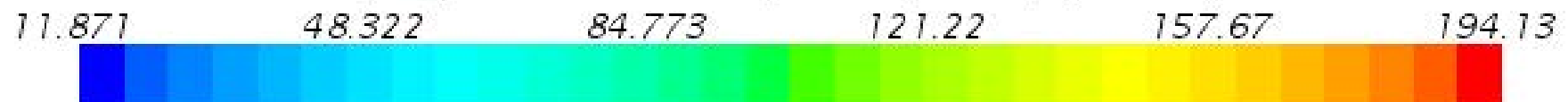


Specified y^+ Heat Transfer Coefficient ($y^+ = 100$)



Good

Specified y^+ Heat Transfer Coefficient (W/m^2-K)





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www.cd-adapco.co.jp

info-jp@cd-adapco.com

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